QUANTUM PHYSICS

22.1 Energy and momentum of a photon

Candidates should be able to:

1 understand that electromagnetic radiation has a particulate nature

2 understand that a photon is a quantum of electromagnetic energy

3 recall and use $E = hf$

4 use the electronvolt (eV) as a unit of energy

5 understand that a photon has momentum and that the momentum is given by $p = E / c$

22.2 Photoelectric effect

Candidates should be able to:

1 understand that photoelectrons may be emitted from a metal surface when it is illuminated by electromagnetic radiation

2 understand and use the terms threshold frequency and threshold wavelength

3 explain photoelectric emission in terms of photon energy and work function energy

4 recall and use $hf = \Phi + \frac{1}{2}m{v_{max}}^2$

5 explain why the maximum kinetic energy of photoelectrons is independent of intensity, whereas the photoelectric current is proportional to intensity

22.3 Wave-particle duality

Candidates should be able to:

1 understand that the photoelectric effect provides evidence for a particulate nature of electromagnetic radiation while phenomena such as interference and diffraction provide evidence for a wave nature

2 describe and interpret qualitatively the evidence provided by electron diffraction for the wave nature of particles

3 understand the de Broglie wavelength as the wavelength associated with a moving particle

4 recall and use $\lambda = h / p$

22.4 Energy levels in atoms and line spectra

Candidates should be able to:

1 understand that there are discrete electron energy levels in isolated atoms (e.g. atomic hydrogen)

2 understand the appearance and formation of emission and absorption line spectra

3 recall and use $hf = E_1 - E_2$

1. M/J/07/Q5

(a) **(i)** Explain what is meant by a *photon*.

....

.... [1]

(ii) Show that the photon energy of light of wavelength 350 nm is 5.68×10^{-19} J. [1]

(iii) State the value of the ratio

$$\frac{\text{energy of photon of light of wavelength 700 nm}}{\text{energy of photon of light of wavelength 350 nm}}.$$

ratio = ... [1]

(b) Two beams of monochromatic light have similar intensities. The light in one beam has wavelength 350 nm and the light in the other beam has wavelength 700 nm.

The two beams are incident separately on three different metal surfaces. The work function of each of these surfaces is shown in Fig. 5.1.

metal	work function / eV
tungsten	4.49
magnesium	3.68
potassium	2.26

Fig. 5.1

(i) Explain what is meant by the *work function* of the surface.

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.... [2]

(ii) State which combination, if any, of monochromatic light and metal surface could give rise to photo-electric emission. Give a quantitative explanation of your answer.

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.... [3]

2. O/N/08/Q7

(a) State three pieces of evidence provided by the photoelectric effect for a particulate nature of electromagnetic radiation.

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2. ....

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3. ....

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[3]

(b) **(i)** Briefly describe the concept of a photon.

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.... [2]

(ii) Explain how lines in the emission spectrum of gases at low pressure provide evidence for discrete electron energy levels in atoms.

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.... [2]

(c) Three electron energy levels in atomic hydrogen are represented in Fig. 7.1.

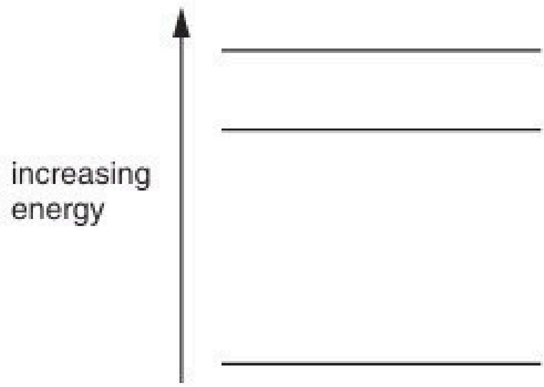

Fig. 7.1

The wavelengths of the spectral lines produced by electron transitions between these three energy levels are 486 nm, 656 nm and 1880 nm.

(i) On Fig. 7.1, draw arrows to show the electron transitions between the energy levels that would give rise to these wavelengths.
Label each arrow with the wavelength of the emitted photon. [3]

(ii) Calculate the maximum change in energy of an electron when making transitions between these levels.

energy = ... J [3]

3. M/J/09/Q8

(a) Explain why, for the photoelectric effect, the existence of a threshold frequency and a very short emission time provide evidence for the particulate nature of electromagnetic radiation, as opposed to a wave theory.

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.... [4]

(b) State and explain two relations in which the Planck constant *h* is the constant of proportionality.

1. ....

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2. ....

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[6]

4. O/N/09/42/Q7

(a) Explain how a line emission spectrum leads to an understanding of the existence of discrete electron energy levels in atoms.

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.... [3]

(b) Some of the lines of the emission spectrum of atomic hydrogen are shown in Fig. 7.1.

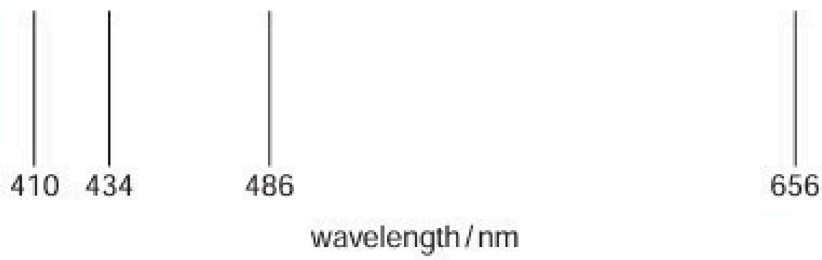

Fig. 7.1

The photon energies associated with some of these lines are shown in Fig. 7.2.

wavelength / nm	photon energy / 10^{-19} J
410	4.85
434	4.58
486	....
656	3.03

Fig. 7.2

(i) Complete Fig. 7.2 by calculating the photon energy for a wavelength of 486 nm.

(ii) Energy levels of a single electron in a hydrogen atom are shown in Fig. 7.3.

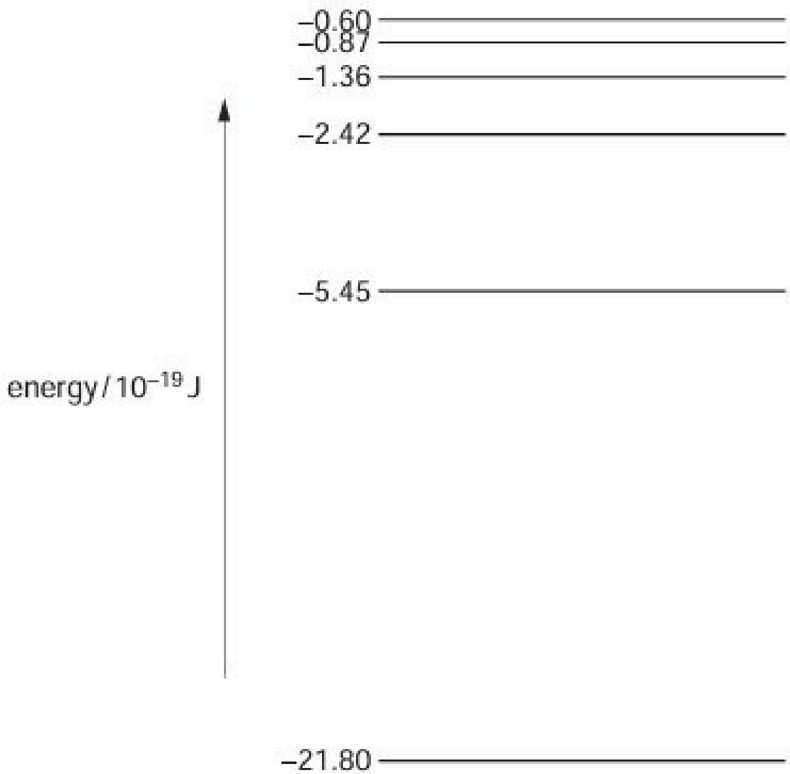

Fig. 7.3 (not to scale)

Use data from **(i)** to show, on Fig. 7.3, the transitions associated with each of the four spectral lines shown in Fig. 7.1. Show each transition with an arrow. [2]

5. O/N/10/41/Q7

(a) State an effect, one in each case, that provides evidence for

(i) the wave nature of a particle,

.... [1]

(ii) the particulate nature of electromagnetic radiation.

.... [1]

(b) Four electron energy levels in an atom are shown in Fig. 7.1.

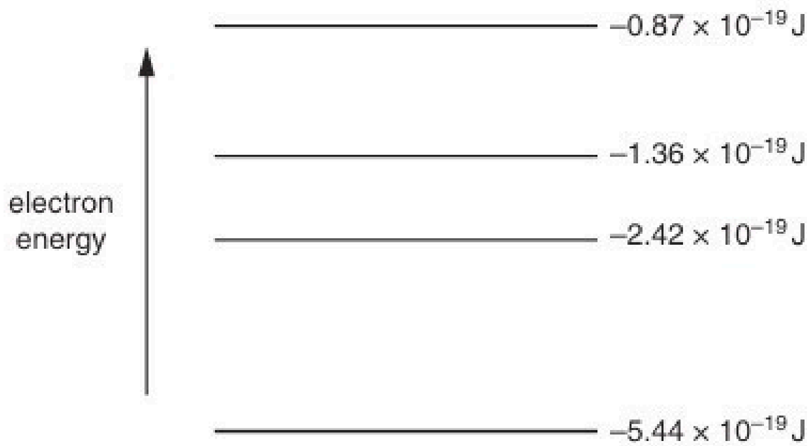

Fig. 7.1 (not to scale)

An emission spectrum is associated with the electron transitions between these energy levels.
For this spectrum,

(i) state the number of lines,

.... [1]

(ii) calculate the minimum wavelength.

wavelength = ... m [2]

6. O/N/10/43/Q8

(a) By reference to the photoelectric effect, state what is meant by the *threshold frequency*.

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.... [2]

(b) The surface of a zinc plate has a work function of 5.8×10^{-19} J.
In a particular laboratory experiment, ultraviolet light of wavelength 120 nm is incident on the zinc plate. A photoelectric current I is detected.
In order to view the apparatus more clearly, a second lamp emitting light of wavelength 450 nm is switched on. No change is made to the ultraviolet lamp.

Using appropriate calculations, state and explain the effect on the photoelectric current of switching on this second lamp.

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.... [4]

7. M/J/11/41/Q7

Experiments are conducted to investigate the photoelectric effect.

(a) It is found that, on exposure of a metal surface to light, either electrons are emitted immediately or they are not emitted at all.

Suggest why this observation does not support a wave theory of light.

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....[3]

(b) Data for the wavelength λ of the radiation incident on the metal surface and the maximum kinetic energy E_K of the emitted electrons are shown in Fig. 7.1.

λ / nm	E_K / 10^{-19} J
650	–
240	4.44

Fig. 7.1

(i) Without any calculation, suggest why no value is given for E_K for radiation of wavelength 650 nm.

....

....[1]

(ii) Use data from Fig. 7.1 to determine the work function energy of the surface.

work function energy = ... J [3]

(c) Radiation of wavelength 240 nm gives rise to a maximum photoelectric current I. The intensity of the incident radiation is maintained constant and the wavelength is now reduced.

State and explain the effect of this change on

(i) the maximum kinetic energy of the photoelectrons,

....

....

....[2]

(ii) the maximum photoelectric current I.

....

....

....[2]

8. M/J/11/42/Q7

(a) State what is meant by the *de Broglie wavelength*.

....

....

.... [2]

(b) An electron is accelerated in a vacuum from rest through a potential difference of 850 V.

(i) Show that the final momentum of the electron is 1.6×10^{-23} N s.

[2]

(ii) Calculate the de Broglie wavelength of this electron.

wavelength = ... m [2]

(c) Describe an experiment to demonstrate the wave nature of electrons.
You may draw a diagram if you wish.

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.... [5]

9. O/N/11/41/Q7

(a) Explain how the line spectrum of hydrogen provides evidence for the existence of discrete electron energy levels in atoms.

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....[3]

(b) Some electron energy levels in atomic hydrogen are illustrated in Fig. 7.1.

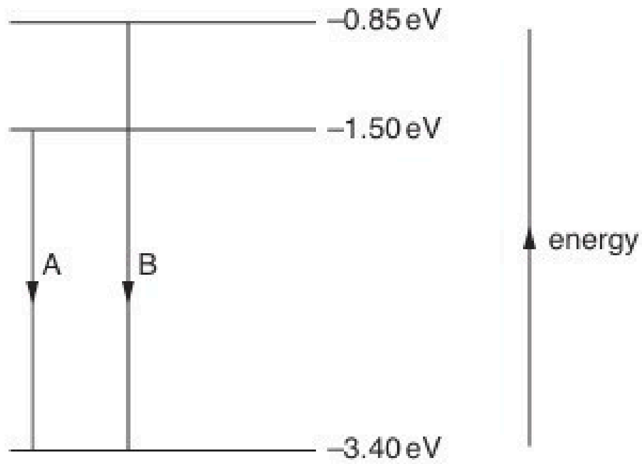

Fig. 7.1

Two possible electron transitions A and B giving rise to an emission spectrum are shown.
These electron transitions cause light of wavelengths 654 nm and 488 nm to be emitted.

(i) On Fig. 7.1, draw an arrow to show a third possible transition. [1]

(ii) Calculate the wavelength of the emitted light for the transition in **(i)**.

wavelength = ... m [3]

(c) The light in a beam has a continuous spectrum of wavelengths from 400 nm to 700 nm. The light is incident on some cool hydrogen gas, as illustrated in Fig. 7.2.

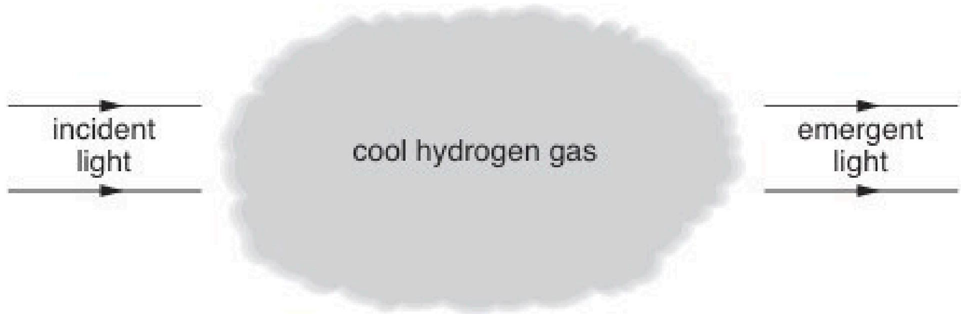

Fig. 7.2

Using the values of wavelength in **(b)**, state and explain the appearance of the spectrum of the emergent light.

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.... [4]

10. O/N/11/43/Q7

An explanation of the photoelectric effect includes the terms photon energy and work function energy.

(a) Explain what is meant by

(i) a *photon*,

....

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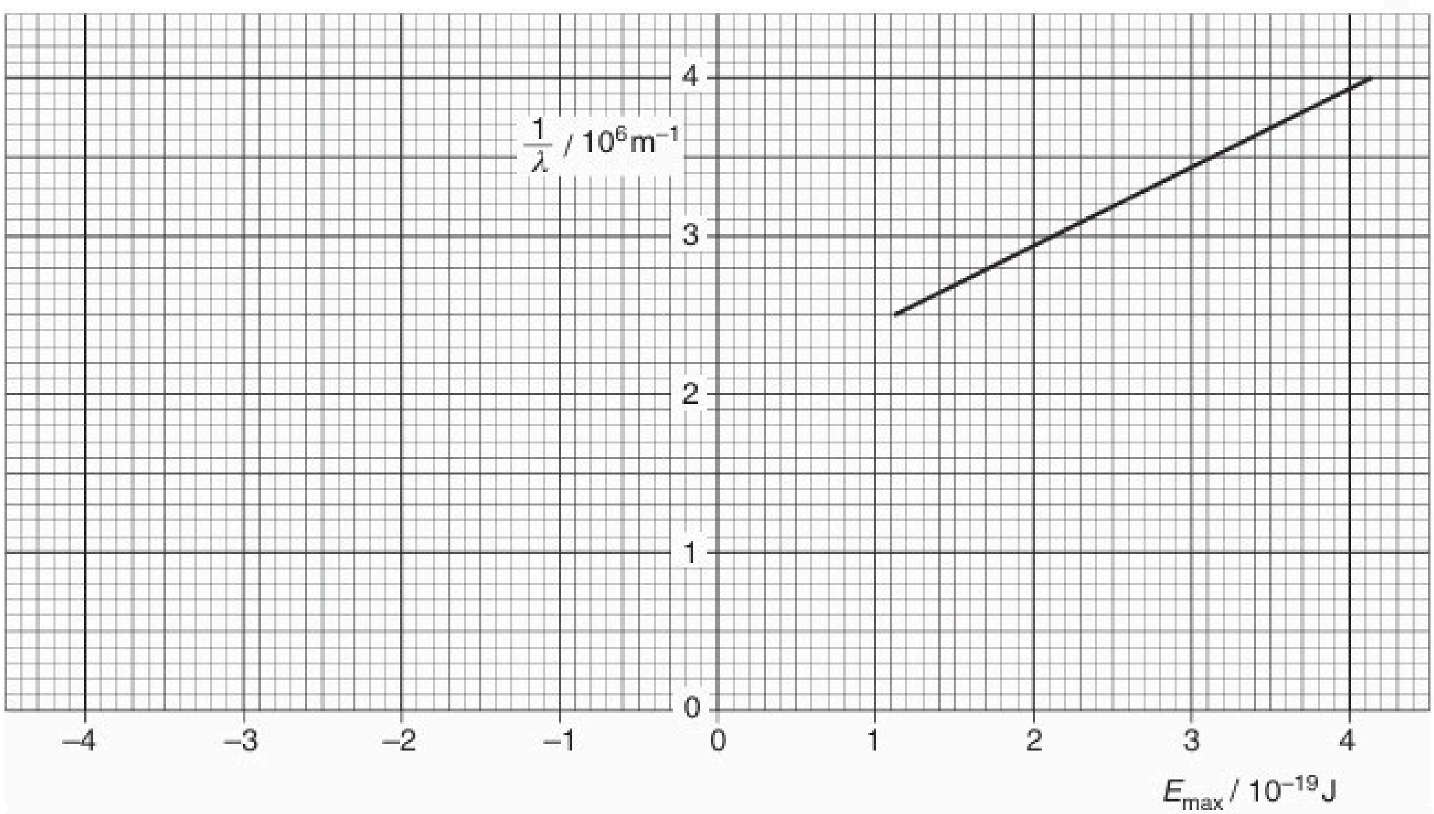
.... [2]

(ii) *work function energy*.

....

.... [1]

(b) In an experiment to investigate the photoelectric effect, a student measures the wavelength λ of the light incident on a metal surface and the maximum kinetic energy E_{max} of the emitted electrons. The variation with E_{max} of $\frac{1}{\lambda}$ is shown in Fig. 7.1.



(i) The work function energy of the metal surface is Φ.
State an equation, in terms of λ, Φ and E_{max}, to represent conservation of energy for the photoelectric effect. Explain any other symbols you use.

....

....

.... [2]

(ii) Use your answer in **(i)** and Fig. 7.1 to determine

1. the work function energy Φ of the metal surface,

Φ = ... J [2]

2. a value for the Planck constant.

Planck constant = ... J s [3]

11. M/J/12/42/Q7

7 The photoelectric effect may be represented by the equation

photon energy = work function energy + maximum kinetic energy of electron.

(a) State what is meant by *work function energy*.

....

.... [1]

(b) The variation with frequency f of the maximum kinetic energy E_K of photoelectrons emitted from the surface of sodium metal is shown in Fig. 7.1.

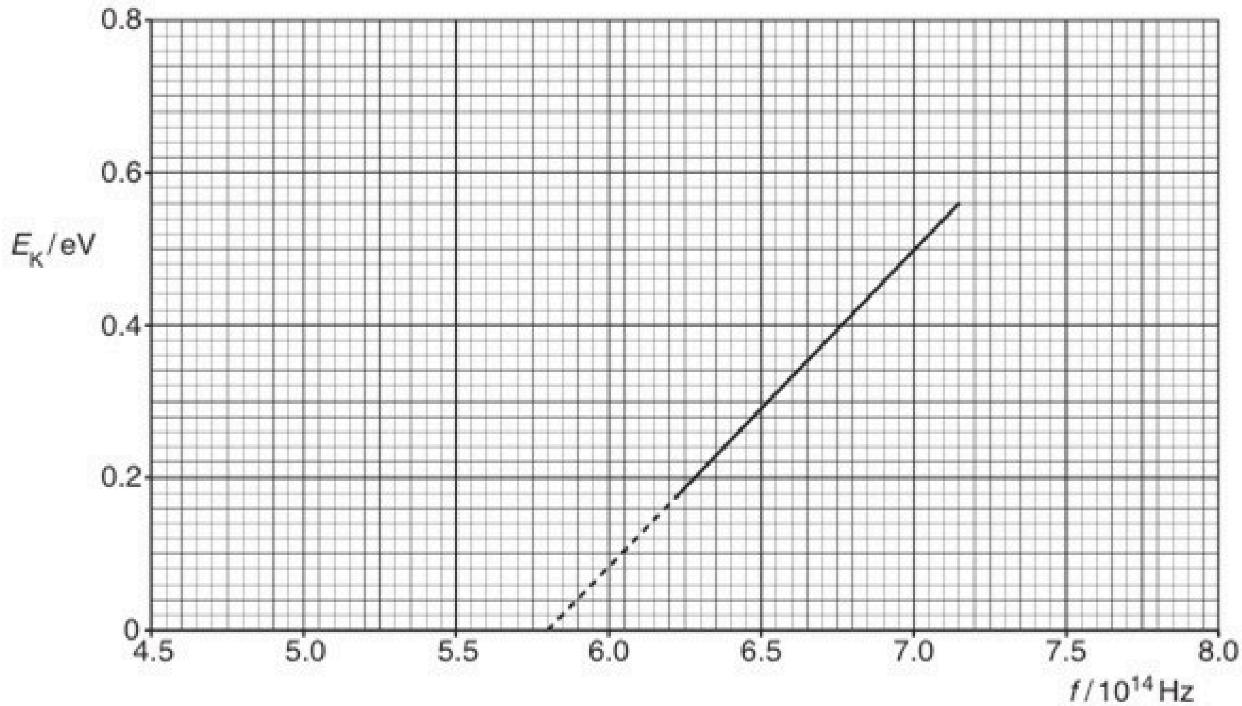


Fig. 7.1

Use the gradient of the graph of Fig. 7.1 to determine a value for the Planck constant h. Show your working.

h = ... J s [2]

(c) The sodium metal in **(b)** has a work function energy of 2.4 eV. The sodium is replaced by calcium which has a work function energy of 2.9 eV.

On Fig. 7.1, draw a line to show the variation with frequency f of the maximum kinetic energy E_K of photoelectrons emitted from the surface of calcium. [3]

12. O/N/12/41/Q8

(a) State what is meant by a *photon*.

....

....

....[2]

(b) It has been observed that, where photoelectric emission of electrons takes place, there is negligible time delay between illumination of the surface and emission of an electron.

State three other pieces of evidence provided by the photoelectric effect for the particulate nature of electromagnetic radiation.

1. ....

....

2. ....

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3. ....

....
[3]

(c) The work function of a metal surface is 3.5 eV. Light of wavelength 450 nm is incident on the surface.
Determine whether electrons will be emitted, by the photoelectric effect, from the surface.

[3]

13. O/N/12/43/Q7

(a) State what is meant by the *de Broglie wavelength*.

....

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.... [2]

(b) An electron is accelerated from rest in a vacuum through a potential difference of 4.7 kV.

(i) Calculate the de Broglie wavelength of the accelerated electron.

wavelength = ... m [5]

(ii) By reference to your answer in **(i)**, suggest why such electrons may assist with an understanding of crystal structure.

....

....

....

.... [2]

14. M/J/13/41/Q7

Some data for the work function energy Φ and the threshold frequency f_0 of some metal surfaces are given in Fig. 7.1.

metal	$\Phi / 10^{-19}$ J	$f_0 / 10^{14}$ Hz
sodium	3.8	5.8
zinc	5.8	8.8
platinum	9.0	

Fig. 7.1

(a) **(i)** State what is meant by the *threshold frequency*.

....

....

.... [2]

(ii) Calculate the threshold frequency for platinum.

threshold frequency = ... Hz [2]

(b) Electromagnetic radiation having a continuous spectrum of wavelengths between 300 nm and 600 nm is incident, in turn, on each of the metals listed in Fig. 7.1. Determine which metals, if any, will give rise to the emission of electrons.

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.... [2]

(c) When light of a particular intensity and frequency is incident on a metal surface, electrons are emitted.
State and explain the effect, if any, on the rate of emission of electrons from this surface for light of the same intensity and higher frequency.

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15. M/J/13/42/Q7

(a) The emission spectrum of atomic hydrogen consists of a number of discrete wavelengths. Explain how this observation leads to an understanding that there are discrete electron energy levels in atoms.

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....

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.... [2]

(b) Some electron energy levels in atomic hydrogen are illustrated in Fig. 7.1.

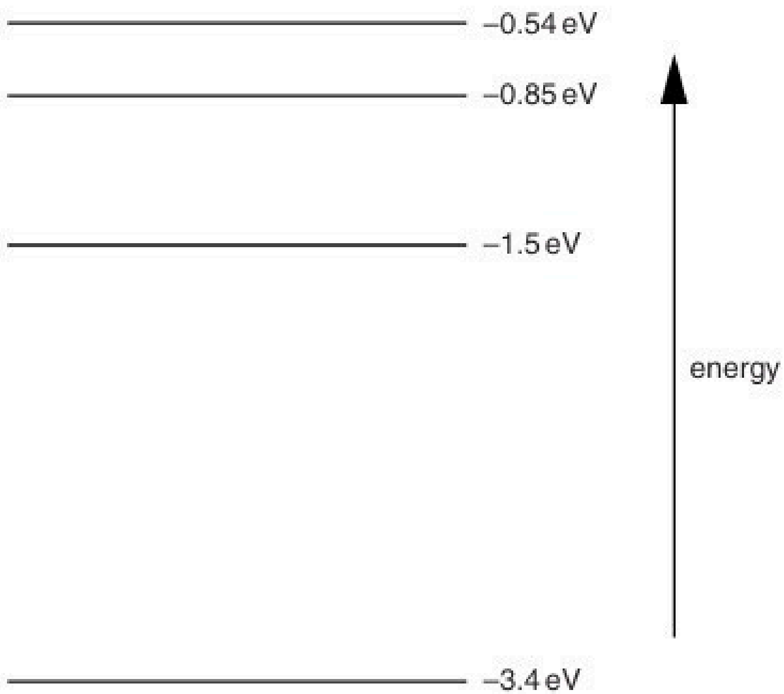

Fig. 7.1

The longest wavelength produced as a result of electron transitions between two of the energy levels shown in Fig. 7.1 is 4.0×10^{-6} m.

(i) On Fig. 7.1,

1. draw, and mark with the letter L, the transition giving rise to the wavelength of 4.0×10^{-6} m, [1]

2. draw, and mark with the letter S, the transition giving rise to the shortest wavelength. [1]

(ii) Calculate the wavelength for the transition you have shown in **(i) part 2**.

wavelength = ... m [3]

(c) Photon energies in the visible spectrum vary between approximately 3.66 eV and 1.83 eV.

Determine the energies, in eV, of photons in the visible spectrum that are produced by transitions between the energy levels shown in Fig. 7.1.

photon energies .... eV [2]

16. O/N/13/41/Q7

Electrons, travelling at speed v in a vacuum, are incident on a very thin carbon film, as illustrated in Fig. 7.1.

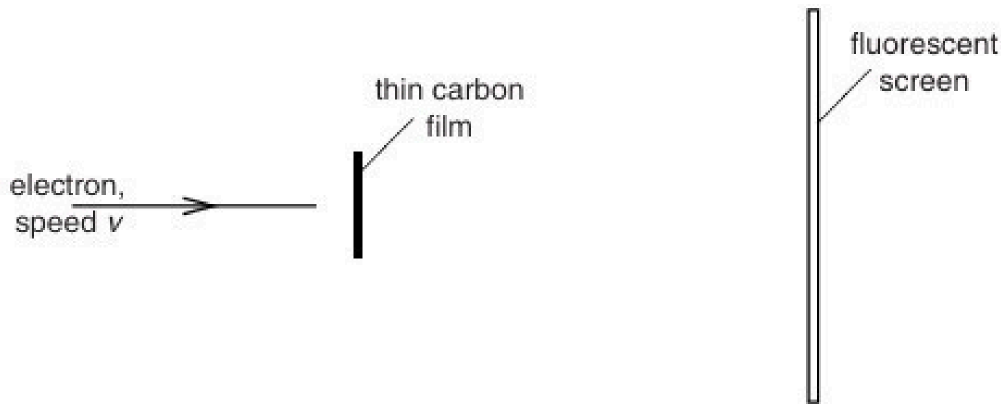

Fig. 7.1

The emergent electrons are incident on a fluorescent screen.
A series of concentric rings is observed on the screen.

(a) Suggest why the observed rings provide evidence for the wave nature of particles.

....

....

....

....[2]

(b) The initial speed of the electrons is increased. State and explain the effect, if any, on the radii of the rings observed on the screen.

....

....

....

....[3]

(c) A proton and an electron are each accelerated from rest through the same potential difference.
Determine the ratio

$$\frac{\text{de Broglie wavelength of the proton}}{\text{de Broglie wavelength of the electron}}.$$

ratio = ... [4]

17. O/N/13/43/Q7

(a) By reference to the photoelectric effect, explain

(i) what is meant by *work function energy*,

....

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.... [2]

(ii) why, even when the incident light is monochromatic, the emitted electrons have a range of kinetic energy up to a maximum value.

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.... [2]

(b) Electromagnetic radiation of frequency f is incident on a metal surface. The variation with frequency f of the maximum kinetic energy E_{MAX} of electrons emitted from the surface is shown in Fig. 7.1.

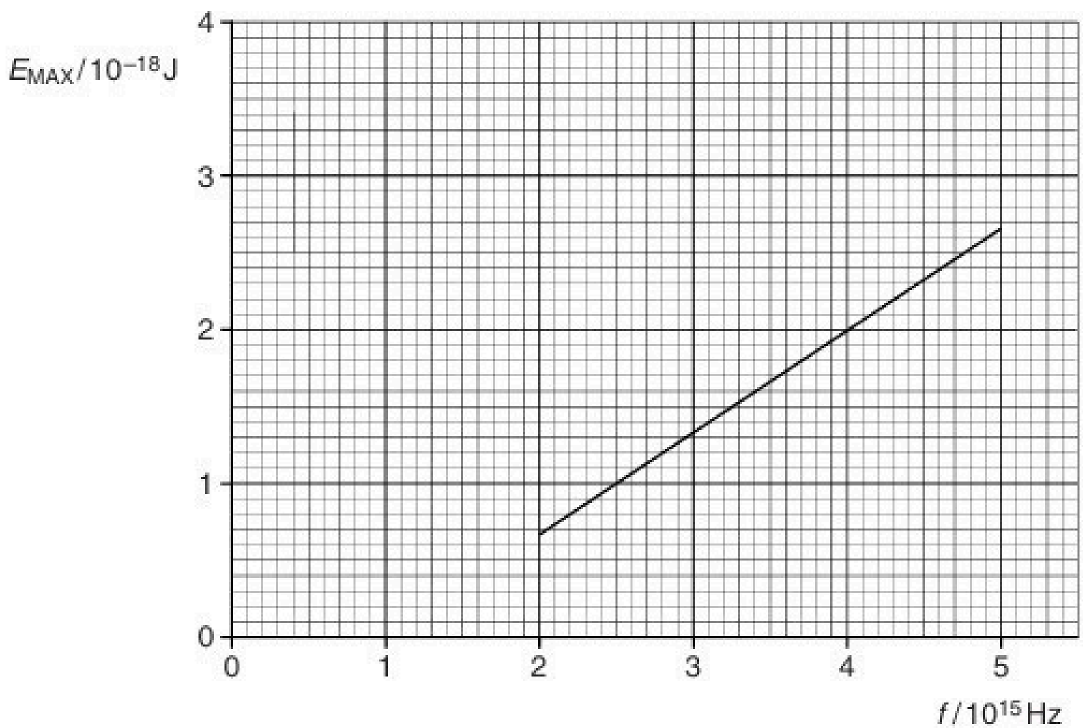

Fig. 7.1

(i) Use Fig. 7.1 to determine the work function energy of the metal surface.

work function energy = ... J [3]

(ii) A second metal has a greater work function energy than that in **(i)**.
On Fig. 7.1, draw a line to show the variation with f of E_{MAX} for this metal. [2]

(iii) Explain why the graphs in **(i)** and **(ii)** do not depend on the intensity of the incident radiation.

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....[2]

18. M/J/14/41/Q9

For a particular metal surface, it is observed that there is a minimum frequency of light below which photoelectric emission does not occur. This observation provides evidence for a particulate nature of electromagnetic radiation.

(a) State three further observations from photoelectric emission that provide evidence for a particulate nature of electromagnetic radiation.

1. ....

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2. ....

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3. ....

.... [3]

(b) Some data for the variation with frequency f of the maximum kinetic energy E_{MAX} of electrons emitted from a metal surface are shown in Fig. 9.1.

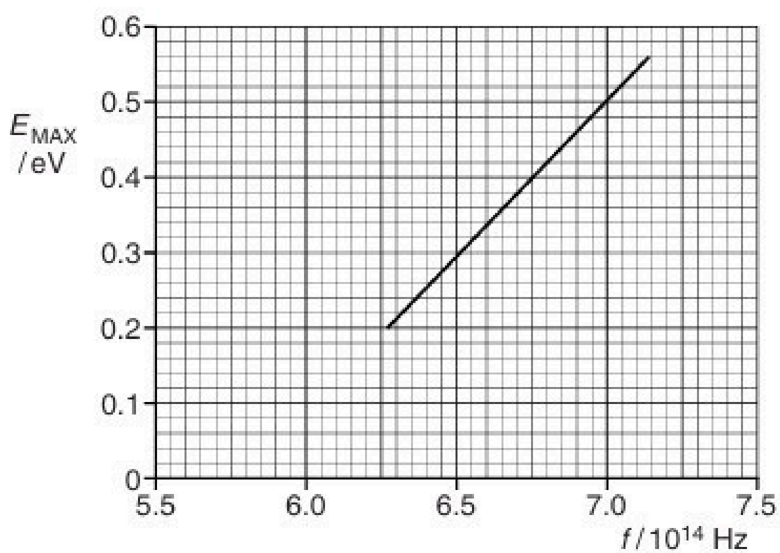

Fig. 9.1

(i) Explain why emitted electrons may have kinetic energy less than the maximum at any particular frequency.

....

....

.... [2]

(ii) Use Fig. 9.1 to determine

1. the threshold frequency,

threshold frequency = ... Hz [1]

2. the work function energy, in eV, of the metal surface.

work function energy = ... eV [3]

19. M/J/14/42/Q8

Light of wavelength 590 nm is incident normally on a surface, as illustrated in Fig. 8.1.

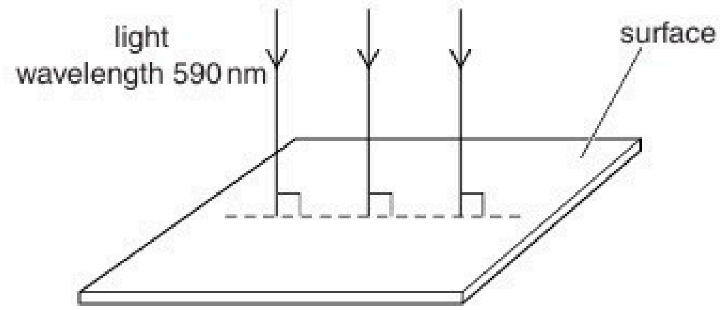

Fig. 8.1

The power of the light is 3.2 mW. The light is completely absorbed by the surface.

(a) Calculate the number of photons incident on the surface in 1.0 s.

number = ... [3]

(b) Use your answer in **(a)** to determine

(i) the total momentum of the photons arriving at the surface in 1.0 s,

momentum = ... $kg\,m\,s^{-1}$ [3]

(ii) the force exerted on the surface by the light.

force = ... N [1]

20. *O/N/14/41/Q8*

White light is incident on a cloud of cool hydrogen gas, as illustrated in Fig. 8.1.

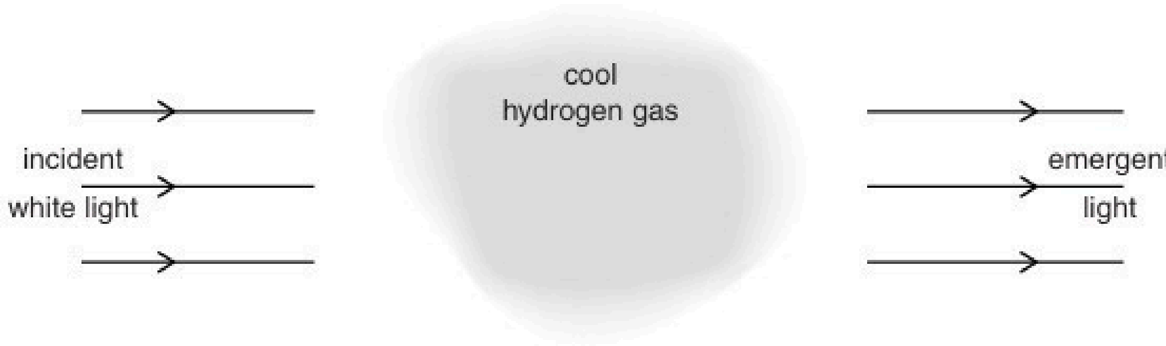

Fig. 8.1

The spectrum of the light emerging from the gas cloud is found to contain a number of dark lines.

(a) Explain why these dark lines occur.

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.... [3]

(b) Some electron energy levels in a hydrogen atom are illustrated in Fig. 8.2.

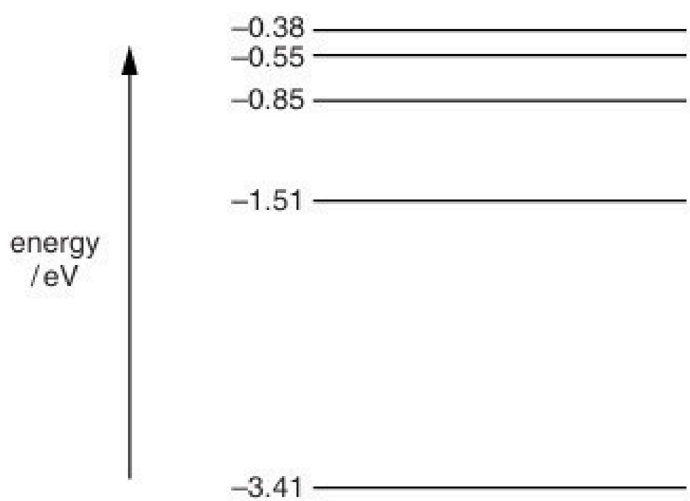

Fig. 8.2

One dark line is observed at a wavelength of 435 nm.

(i) Calculate the energy, in eV, of a photon of light of wavelength 435 nm.

energy = ... eV [4]

(ii) On Fig. 8.2, draw an arrow to indicate the energy change that gives rise to this dark line. [1]

21. O/N/14/43/Q8

(a) State what is meant by a *photon*.

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.... [2]

(b) A beam of light is incident normally on a metal surface, as illustrated in Fig. 8.1.

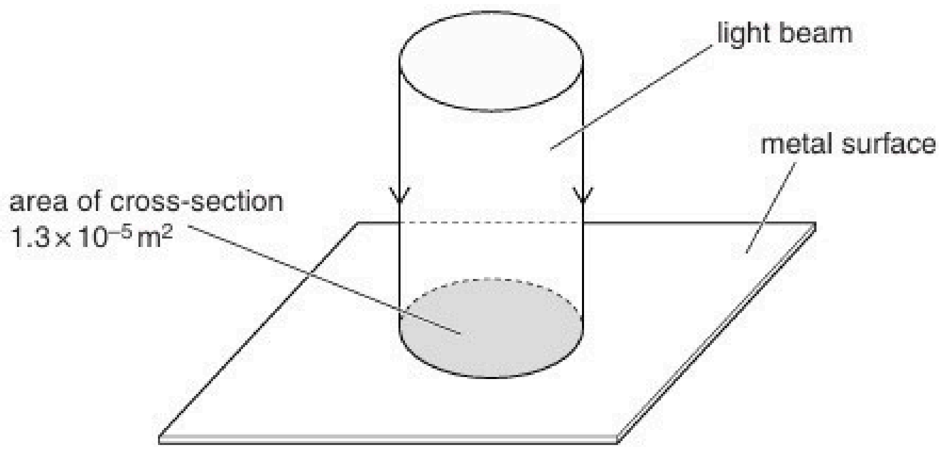

Fig. 8.1

The beam of light has cross-sectional area $1.3 \times 10^{-5}\,m^2$ and power $2.7 \times 10^{-3}\,W$.
The light has wavelength 570 nm.

The light energy is absorbed by the metal and no light is reflected.

(i) Show that a photon of this light has an energy of $3.5 \times 10^{-19}\,J$.

[1]

(ii) Calculate, for a time of 1.0 s,

1. the number of photons incident on the surface,

number = ... [2]

2. the change in momentum of the photons.

change in momentum = ... $kg\,m\,s^{-1}$ [3]

(c) Use your answer in **(b)(ii)** to calculate the pressure that the light exerts on the metal surface.

pressure = ... Pa [2]

22. M/J/15/41/Q8

A photon of wavelength 6.50×10^{-12} m is incident on an isolated stationary electron, as illustrated in Fig. 8.1.

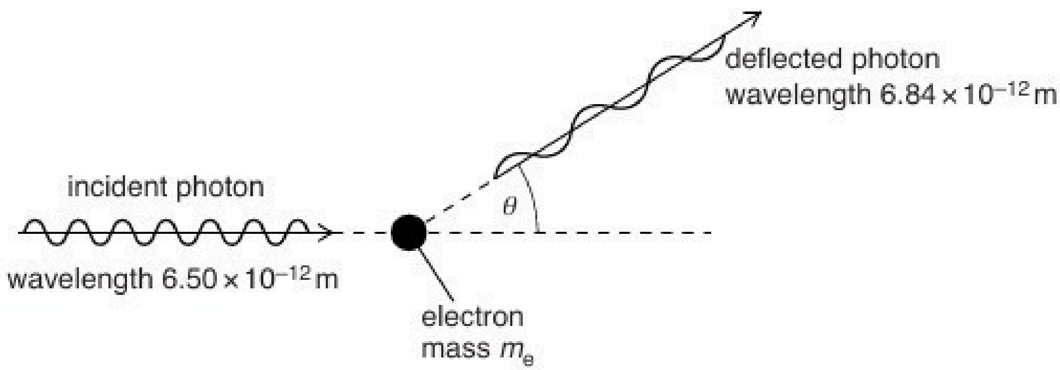

Fig. 8.1

The photon is deflected elastically by the electron of mass m_e. The wavelength of the deflected photon is 6.84×10^{-12} m.

(a) Calculate, for the incident photon,

(i) its momentum,

momentum = ... Ns [2]

(ii) its energy.

energy = ...J [2]

(b) The angle θ through which the photon is deflected is given by the expression

$$\Delta\lambda = \frac{h}{m_e c}(1 - \cos\theta)$$

where $\Delta\lambda$ is the change in wavelength of the photon, h is the Planck constant and c is the speed of light in free space.

(i) Calculate the angle θ.

θ = ... ° [2]

(ii) Use energy considerations to suggest why $\Delta\lambda$ must always be positive.

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.... [3]

23. M/J/15/42/Q6

(a) Explain what is meant by a *photon*.

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....[2]

(b) An X-ray photon of energy 3.06×10^{-14} J is incident on an isolated stationary electron, as illustrated in Fig. 6.1.

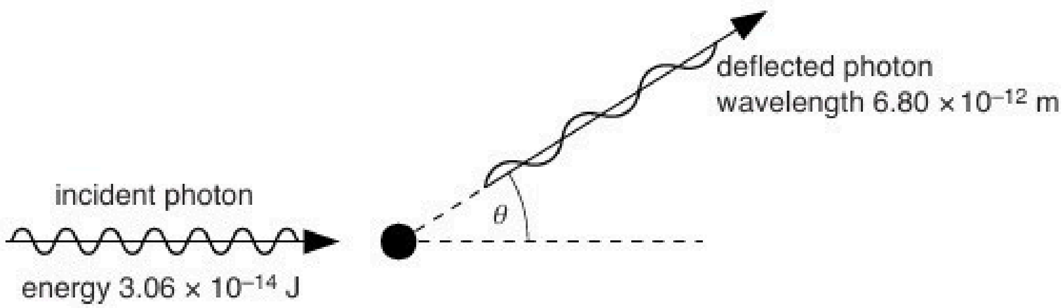

Fig. 6.1

The photon is deflected elastically by the electron through angle θ. The deflected photon has a wavelength of 6.80×10^{-12} m.

(i) On Fig. 6.1, draw an arrow to indicate a possible initial direction of motion of the electron after the photon has been deflected. [1]

(ii) Calculate

1. the energy of the deflected photon,

photon energy = ... J [2]

2. the speed of the electron after the photon has been deflected.

speed = ... $m s^{-1}$ [3]

(c) Explain why the magnitude of the final momentum of the electron is not equal to the change in magnitude of the momentum of the photon.

....

....

.... [2]

24. O/N/15/41/Q7

(a) By reference to the photoelectric effect, state what is meant by the *threshold frequency*.

....

....

....[2]

(b) Electrons are emitted from a metal surface when light of a particular wavelength is incident on the surface.
Explain why the emitted electrons have a range of values of kinetic energy below a maximum value.

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....[2]

(c) The wavelength of the incident radiation is λ.
The variation with $1/\lambda$ of the maximum kinetic energy E_{MAX} of electrons emitted from a metal surface is shown in Fig. 7.1.

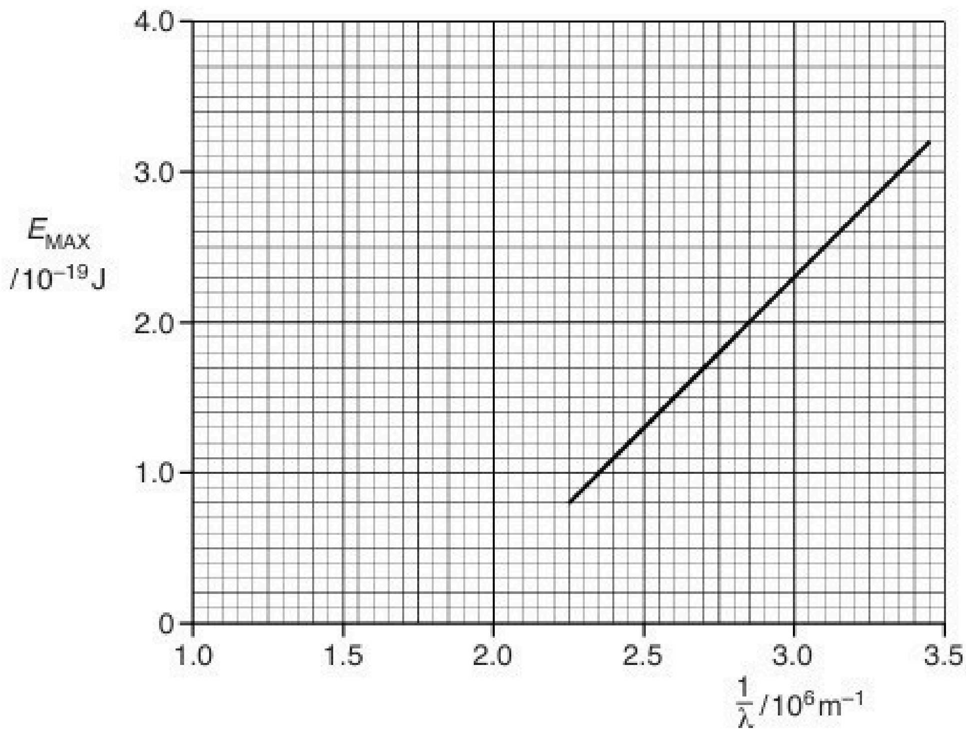

Fig. 7.1

(i) Use Fig. 7.1 to determine, without reference to the work function energy, the threshold frequency f_0.

f_0 = ... Hz [2]

(ii) Use your answer in **(i)** to calculate the work function energy Φ.

Φ = ... J [2]

(d) Caesium metal has a work function energy of 2.2×10^{-19} J.

On the axes of Fig. 7.1, sketch a graph to show the variation with $1/\lambda$ of E_{MAX} for caesium metal. [2]

25. O/N/15/43/Q8

Light of wavelength λ is incident on a metal surface having a work function energy Φ.

Photoelectrons of maximum kinetic energy E_{MAX} are emitted from the surface.

(a) State an equation relating Φ, E_{MAX} and λ. Explain any other symbols you use.

....

....

.... [2]

(b) The variation with $1/\lambda$ of E_{MAX} is shown in Fig. 8.1.

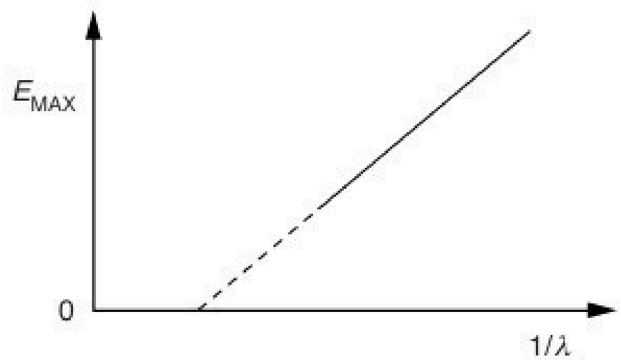

Fig. 8.1

(i) By reference to your answer in **(a)**, explain why the gradient of the line does not depend on the metal surface.

....

....

.... [2]

(ii) The work function energy of sodium is 2.28 eV.

Determine the minimum wavelength λ_0 at which E_{MAX} is zero.

λ_0 = ... m [3]

26. F/M/16/42/Q11

(a) With reference to the photoelectric effect, state what is meant by the *threshold frequency*.

....

....

....[2]

(b) Electromagnetic radiation of wavelength λ is incident on a metal surface. Electrons of maximum kinetic energy E_{MAX} are emitted.

(i) On Fig. 11.1, sketch the variation with $1/\lambda$ of E_{MAX}.

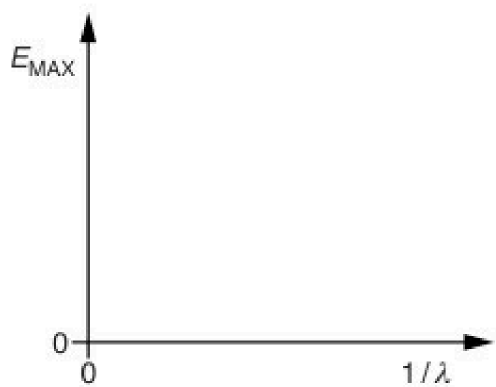

Fig. 11.1

[2]

(ii) State an equation relating the gradient of the graph drawn on Fig. 11.1 to the Planck constant *h*.
Explain any symbols you use.

....

....[1]

(iii) Explain why, for any particular wavelength of electromagnetic radiation, most of the electrons are emitted with kinetic energies less than the maximum value E_{MAX}.

....

....

....

....[2]

(iv) Light of a particular wavelength is incident on a metal surface and gives rise to a photoelectric current.

The wavelength is reduced. The intensity of the light is kept constant.

State and explain the effect, if any, on the photoelectric current.

....

....

....

....

....[3]

[Total: 10]

27. O/N/16/41/Q10

(a) Explain what is meant by the *photoelectric effect*.

....

....

.... [2]

(b) One wavelength of electromagnetic radiation emitted from a mercury vapour lamp is 436 nm.

Calculate the photon energy corresponding to this wavelength.

energy = ... J [2]

(c) Light from the lamp in **(b)** is incident, separately, on the surfaces of caesium and tungsten metal.

Data for the work function energies of caesium and tungsten metal are given in Fig. 10.1.

metal	work function energy / eV
caesium	1.4
tungsten	4.5

Fig. 10.1

Calculate the threshold wavelength for photoelectric emission from

(i) caesium,

threshold wavelength = ... nm [2]

(ii) tungsten.

threshold wavelength = ... nm [1]

(d) Use your answers in **(c)** to state and explain whether the radiation from the mercury lamp of wavelength 436 nm will give rise to photoelectric emission from each of the metals.

caesium: ....

....

tungsten: ....

....

[2]

[Total: 9]

28. O/N/16/42/Q12

(a) State an effect, one in each case, that provides evidence for

(i) the wave nature of a particle,

....[1]

(ii) the particulate nature of electromagnetic radiation.

....[1]

(b) Four electron energy levels in an isolated atom are shown in Fig. 12.1.

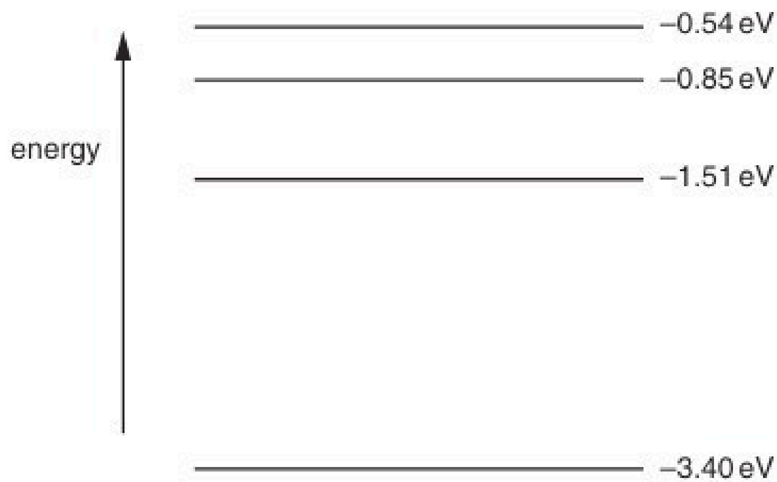

Fig. 12.1

For the emission spectrum associated with these energy levels,

(i) on Fig. 12.1, mark with an arrow the transition that gives rise to the shortest wavelength, [1]

(ii) show that the wavelength of the transition in **(i)** is 4.35×10^{-7} m.

[2]

(c) **(i)** State what is meant by the *de Broglie wavelength*.

....

....

....[2]

(ii) Calculate the speed of an electron having a de Broglie wavelength equal to the wavelength in **(b)(ii)**.

speed = ... $m\,s^{-1}$ [2]

[Total: 9]

29. F/M/17/42/Q10

(a) State what is meant by a *photon*.

....

....[2]

(b) Light in a beam has a continuous spectrum that lies within the visible region. The photons of light have energies ranging from 1.60 eV to 2.60 eV.

The beam passes through some hydrogen gas. It then passes through a diffraction grating and an absorption spectrum is observed.

(i) All of the light absorbed by the hydrogen is re-emitted. Explain why dark lines are still observed in the absorption spectrum.

....

....[1]

(ii) Some of the energy levels of an electron in a hydrogen atom are illustrated in Fig. 10.1.

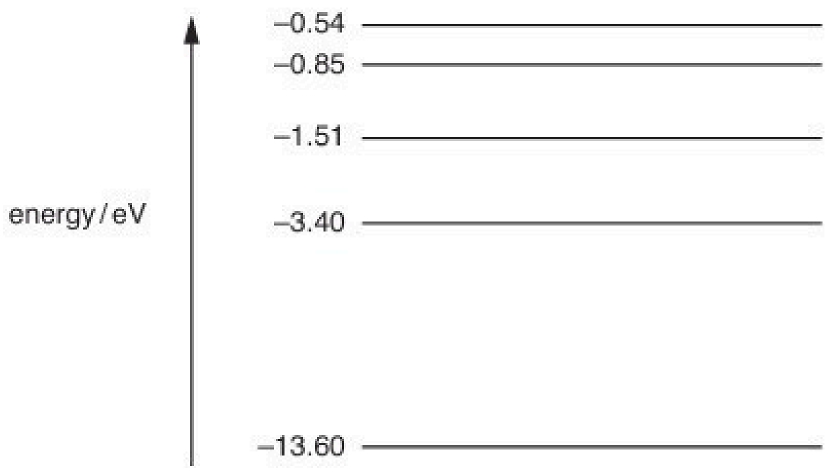

Fig. 10.1 (not to scale)

The dark lines in the absorption spectrum are the result of electron transitions between energy levels.

On Fig. 10.1, draw arrows to show the initial electron transitions between energy levels that could give rise to dark lines in the absorption spectrum. [2]

(iii) Calculate the shortest wavelength of the light in the beam.

wavelength = ...m [3]

30. M/J/17/41/Q11

1 A beam of light consists of a continuous range of wavelengths from 420 nm to 740 nm.
The light passes through a cloud of cool gas, as shown in Fig. 11.1.

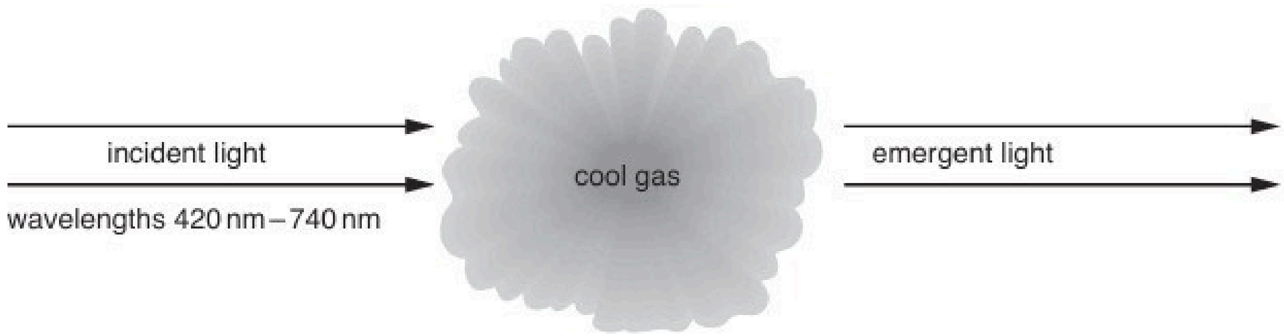

Fig. 11.1

(a) The spectrum of the light emerging from the cloud of cool gas is viewed using a diffraction grating.
Explain why this spectrum contains a number of dark lines.

....

....

....

....

....

....[4]

(b) Some of the electron energy levels of the atoms in the cloud of gas are represented in Fig. 11.2.

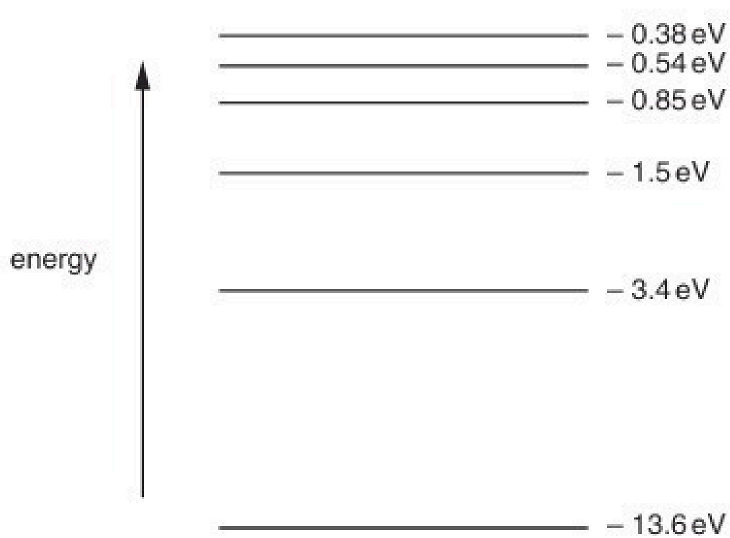

Fig. 11.2 (not to scale)

(i) Light of wavelength 420 nm has a photon energy of 2.96 eV.
Calculate the photon energy, in eV, of light of wavelength 740 nm.

photon energy = ... eV [2]

(ii) Use data from **(i)** and your answer in **(i)** to show, on Fig. 11.2, the changes in energy levels giving rise to the dark lines in **(a)**. [2]

[Total: 8]

31. M/J/17/42/Q10

(a) Briefly describe two phenomena associated with the photoelectric effect that cannot be explained using a wave theory of light.

1. ....

....

2. ....

....
[2]

(b) The maximum energy E_{MAX} of electrons emitted from a metal surface when illuminated by light of wavelength λ is given by the expression

$$E_{MAX} = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

where h is the Planck constant and c is the speed of light.

(i) Identify the symbol λ_0.

....[1]

(ii) The variation with $\frac{1}{\lambda}$ of E_{MAX} for the metal surface is shown in Fig. 10.1.

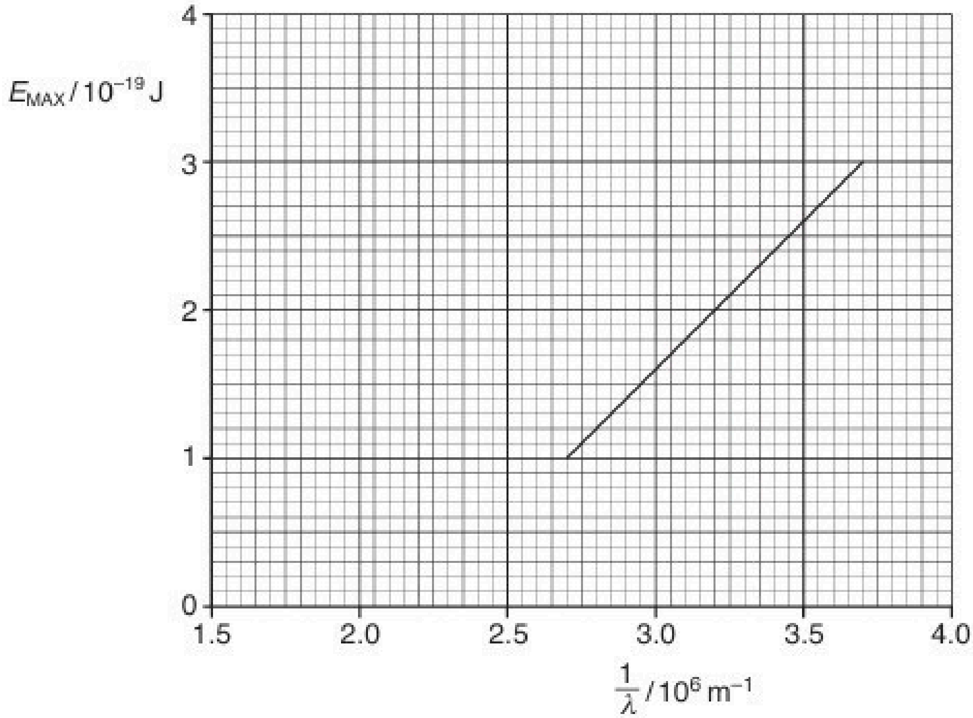

Fig. 10.1

1. Use Fig. 10.1 to determine the magnitude of λ_0.

λ_0 = ... m [1]

2. Use the gradient of Fig. 10.1 to determine a value for the Planck constant *h*.

h = ... J s [3]

(c) The metal surface in **(b)** becomes oxidised.
Photoelectric emission is still observed but the work function energy is increased.

On Fig. 10.1, draw a line to show the variation with $\frac{1}{\lambda}$ of E_{MAX} for the oxidised surface. [2]

[Total: 9]

32. O/N/17/41/Q11

(a) State what is meant by a *photon*.

....

.... [1]

(b) Indium-123 ($^{123}_{49}In$) is radioactive.
A nucleus of indium-123 emits a γ-ray photon of energy 1.1 MeV.

Determine, for this γ-radiation,

(i) the frequency,

frequency = ... Hz [2]

(ii) the momentum of a photon.

momentum = ... N s [2]

(c) The indium-123 nucleus is stationary before emission of the γ-ray photon.
Use your answer in **(b)(ii)** to estimate the recoil speed of the nucleus after emission of the photon.

speed = ... $m s^{-1}$ [2]

33. O/N/17/42/Q10

(a) A metal surface is illuminated with light of a single wavelength λ.
On Fig. 10.1, sketch the variation with λ of the maximum kinetic energy E_{MAX} of the electrons emitted from the surface.
On your graph mark, with the symbol λ_0, the threshold wavelength.

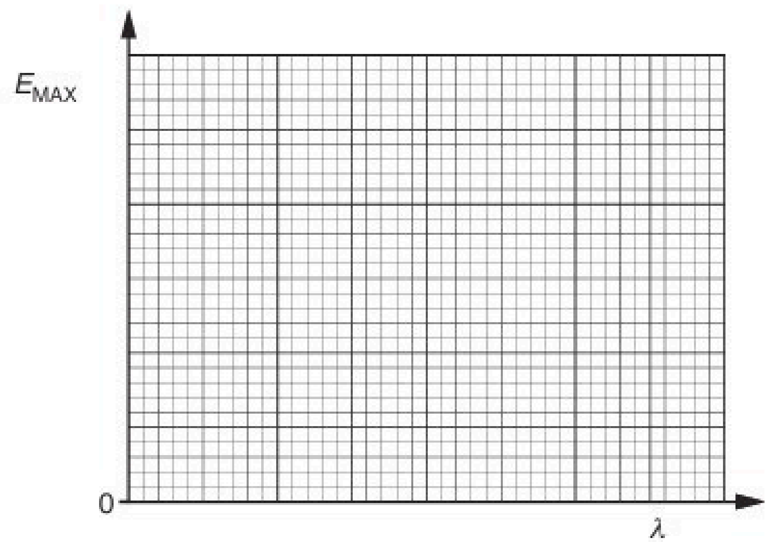

Fig. 10.1 [3]

(b) A neutron is moving in a straight line with momentum p.
The de Broglie wavelength associated with this neutron is λ.
On Fig. 10.2, sketch the variation with momentum p of the de Broglie wavelength λ.

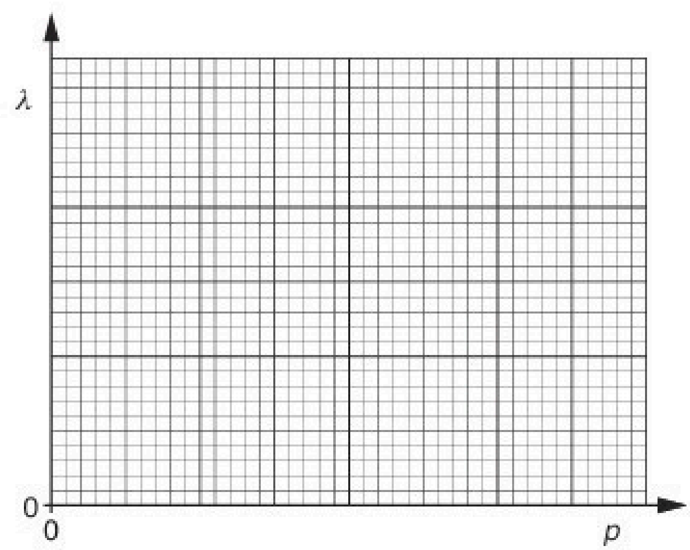

Fig. 10.2 [2]

[Total: 5]

34. M/J/18/42/Q10

(a) Describe the *photoelectric effect*.

....

....

....[2]

(b) Data for the work function energy Φ of two metals are shown in Fig. 10.1.

	Φ / J
sodium	3.8×10^{-19}
zinc	5.8×10^{-19}

Fig. 10.1

Light of wavelength 420 nm is incident on the surface of each of the metals.

(i) State what is meant by a *photon*.

....

....

....[2]

(ii) Calculate the energy of a photon of the incident light.

energy = ... J [2]

(iii) State whether photoelectric emission will occur from each of the metals.

sodium: ....

zinc: ....

[1]

35. O/N/18/41/Q11

A stationary isolated nucleus emits a γ-ray photon of energy 0.51 MeV.

(a) State what is meant by a *photon*.

....

....

....[2]

(b) For the γ-ray photon, calculate

(i) its wavelength,

wavelength = ... m [2]

(ii) its momentum.

momentum = ...N s [2]

(c) **(i)** For this nucleus, determine the change in mass Δm during the decay that gives rise to the energy of the γ-ray photon.

Δm = ... kg [2]

(ii) Explain why, after the decay, the nucleus is no longer stationary.

....

....

....[1]

[Total: 9]

36. O/N/18/42/Q11

(a) State what is meant by a *photon*.

....

....

....[2]

(b) Describe the appearance of a visible line emission spectrum, as seen using a diffraction grating.

....

....

....

....[2]

(c) The lowest electron energy levels in an isolated hydrogen atom are shown in Fig. 11.1.

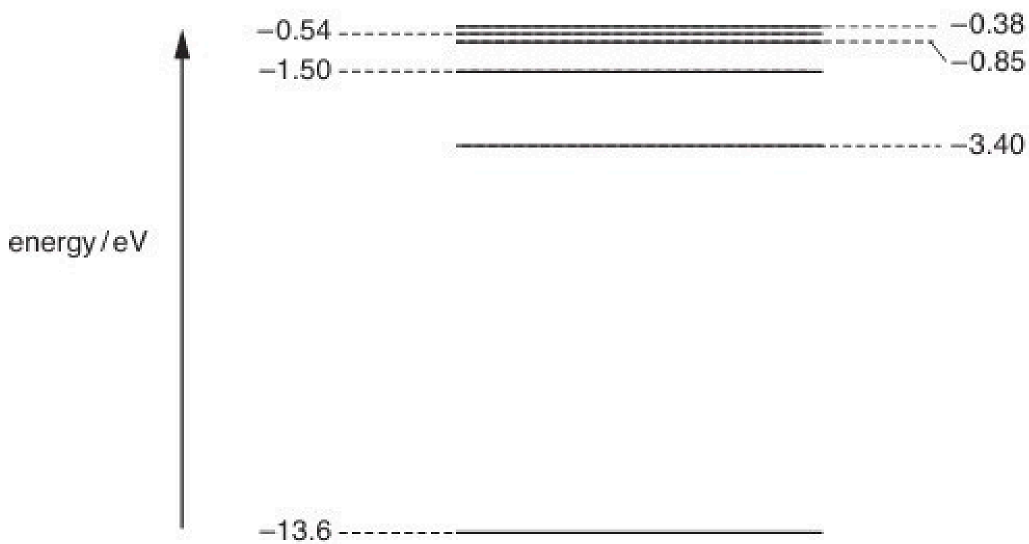

Fig. 11.1 (not to scale)

(i) An electron is initially at the energy level −0.85 eV. State the total number of different wavelengths that may be emitted as the electron de-excites (loses energy).

number = ... [1]

(ii) Photons resulting from electron de-excitation from the −0.85 eV energy level are incident on the surface of a sample of platinum.

Platinum has a work function energy of 5.6 eV.

Determine

1. the maximum kinetic energy, in eV, of a photoelectron emitted from the surface of the platinum,

maximum energy = ... eV [2]

2. the wavelength of the photon producing the photoelectron in **(ii) part 1**.

wavelength = ... m [3]

[Total: 10]

37. M/J/19/41/Q11

(a) State **three** pieces of evidence provided by the photoelectric effect for a particulate nature of electromagnetic radiation.

1. ....

....

2. ....

....

3. ....

.... [3]

(b) The work function energies of some metals are shown in Fig. 11.1.

	work function energy / eV
sodium	2.4
calcium	2.9
zinc	3.6
silver	4.3

Fig. 11.1

Each metal is irradiated with electromagnetic radiation of wavelength 380 nm.

(i) Calculate the energy, in eV, of a photon of electromagnetic radiation of wavelength 380 nm.

energy = ...eV [3]

(ii) Determine which metals will give rise to the emission of photoelectrons. Explain your answer.

....

.... [2]

(c) Photons of wavelength 380 nm are incident normally on a metal surface at a rate of $7.6 \times 10^{14}\,s^{-1}$.
All the photons are absorbed in the surface and no photoelectrons are emitted.

Calculate the force exerted on the metal surface by the incident photons.

force = ... N [3]

[Total: 11]

38. M/J/19/42/Q11

(a) State what is meant by a *photon*.

....

....

....[2]

(b) A stationary cobalt-60 ($^{60}_{27}Co$) nucleus emits a γ-ray photon of energy 1.18 MeV.

(i) Calculate the wavelength of the photon.

wavelength = ... m [2]

(ii) Show that the momentum of the photon is 6.3×10^{-22} N s.

[2]

(c) Use information in **(b)(ii)** to determine the recoil speed of the cobalt-60 nucleus when the γ-ray photon is emitted.

speed = ... m s^{-1} [2]

39. O/N/19/41/Q11

(a) With reference to the photoelectric effect, state what is meant by *work function energy*.

....

....

.... [2]

(b) The work function energy of a clean metal surface is 5.5×10^{-19} J.

Electromagnetic radiation of wavelength 280 nm is incident on the metal surface. The metal is in a vacuum.

(i) Calculate:

1. the photon energy

photon energy = ... J [2]

2. the maximum speed v_{MAX} of the electrons emitted from the surface.

v_{MAX} = ... $m s^{-1}$ [3]

(ii) Explain why most of the emitted electrons will have a speed lower than v_{MAX}.

....

.... [1]

(c) The electromagnetic radiation incident on the metal surface may change in intensity or in frequency.

Complete Fig. 11.1 by inserting either '*increases*' or '*decreases*' or '*no change*' to describe the effects of the changes shown on the maximum speed and on the rate of emission of electrons.

change	maximum speed of electrons	rate of emission of electrons
reduced intensity at constant frequency	....	....
increased frequency at constant intensity	....	....

Fig. 11.1

[4]

[Total: 12]

40. F/M/20/42/Q10

(a) By reference to the photoelectric effect, explain what is meant by *work function energy*.

....

....

.... [2]

(b) In an experiment, electromagnetic radiation of frequency f is incident on a metal surface.

The results in Fig. 10.1 show the variation with frequency f of the maximum kinetic energy E_{MAX} of electrons emitted from the surface.

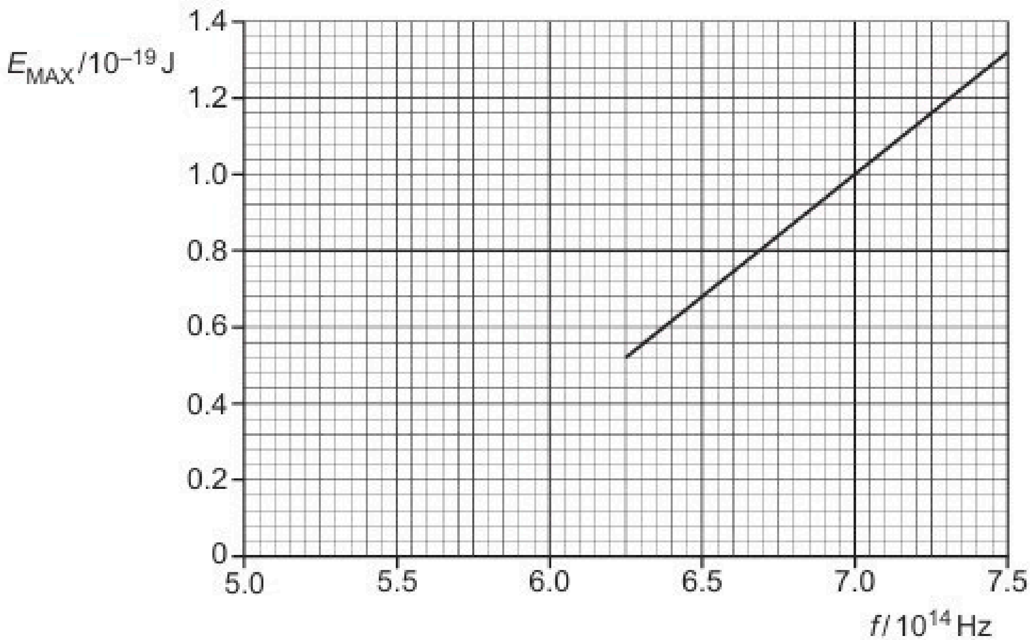

Fig. 10.1

(i) Determine the work function energy in J of the metal used in the experiment.

work function energy = ... J [2]

(ii) The work function energy in eV for some metals is given in Table 10.1.

Table 10.1

metal	work function / eV
tungsten	4.49
magnesium	3.68
potassium	2.26

Determine the metal used in the experiment. Show your working.

....

.... [1]

(c) The intensity of the electromagnetic radiation for one particular frequency in **(b)** is increased.

State and explain the change, if any, in:

(i) the maximum kinetic energy of the emitted electrons

....

.... [1]

(ii) the rate of emission of photoelectrons.

....

.... [1]

[Total: 7]

41. M/J/20/41/Q11

An electron, at rest, has mass m_e and charge $-q$.

A positron is a particle that, at rest, has mass m_e and charge $+q$.

A positron interacts with an electron. The electron and the positron may be considered to be at rest.
The outcome of this interaction is that the electron and the positron become two gamma-ray (γ-ray) photons, each having the same energy.

(a) Calculate, for one of the γ-ray photons:

(i) the photon energy, in J

energy = ... J [2]

(ii) its momentum.

momentum = ... N s [2]

(b) State and explain the direction, relative to each other, in which the γ-ray photons are emitted.

....

....

....

.... [2]

[Total: 6]

42. O/N/20/41/Q11

A photon of wavelength 540 nm collides with an isolated stationary electron, as illustrated in Fig. 11.1.

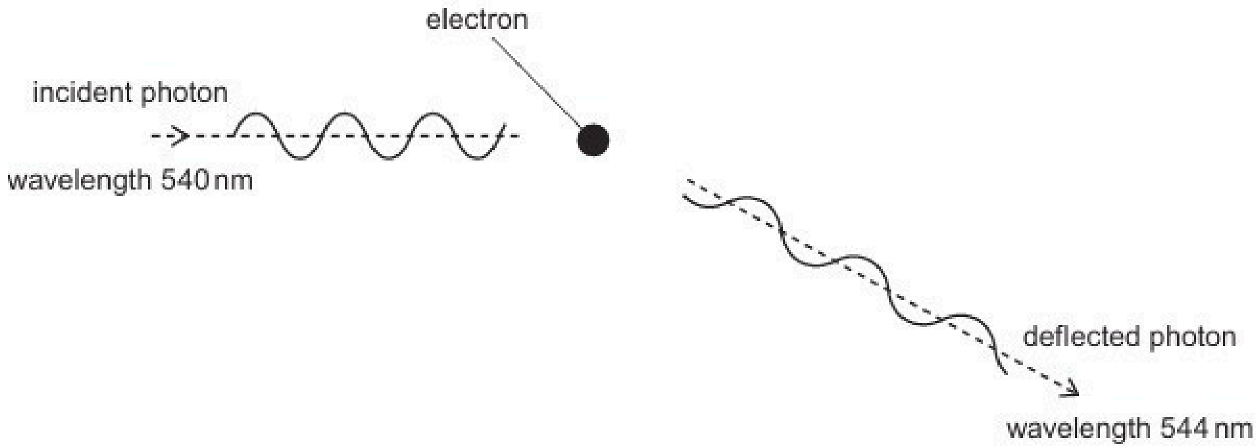

Fig. 11.1

The photon is deflected elastically by the electron.
The wavelength of the deflected photon is 544 nm.

(a) (i) State what is meant by a *photon*.

....

....

.... [2]

(ii) On Fig. 11.1, draw an arrow to indicate the approximate direction of motion of the deflected electron. [1]

(b) Calculate:

(i) the momentum of the deflected photon

momentum = ... N s [2]

(ii) the energy transferred to the deflected electron.

energy = ... J [2]

(c) Another photon of wavelength 540 nm collides with an isolated stationary electron.

Explain why it is not possible for the deflected photon to have a wavelength less than 540 nm.

....

....

.... [2]

[Total: 9]

43. O/N/20/42/Q11

(a) Electromagnetic radiation is incident on a metal surface.

It is observed that there is a minimum frequency of electromagnetic radiation below which emission of electrons does not occur.

This observation provides evidence for a particulate nature of electromagnetic radiation.

State **two** other observations associated with photoelectric emission that provide evidence for a particulate nature of electromagnetic radiation.

1. ....

....

2. ....

....

[2]

(b) The maximum kinetic energy E_{MAX} of electrons emitted from a metal surface is determined for different wavelengths λ of the electromagnetic radiation incident on the surface.

The variation with $\frac{1}{\lambda}$ of E_{MAX} is shown in Fig. 11.1.

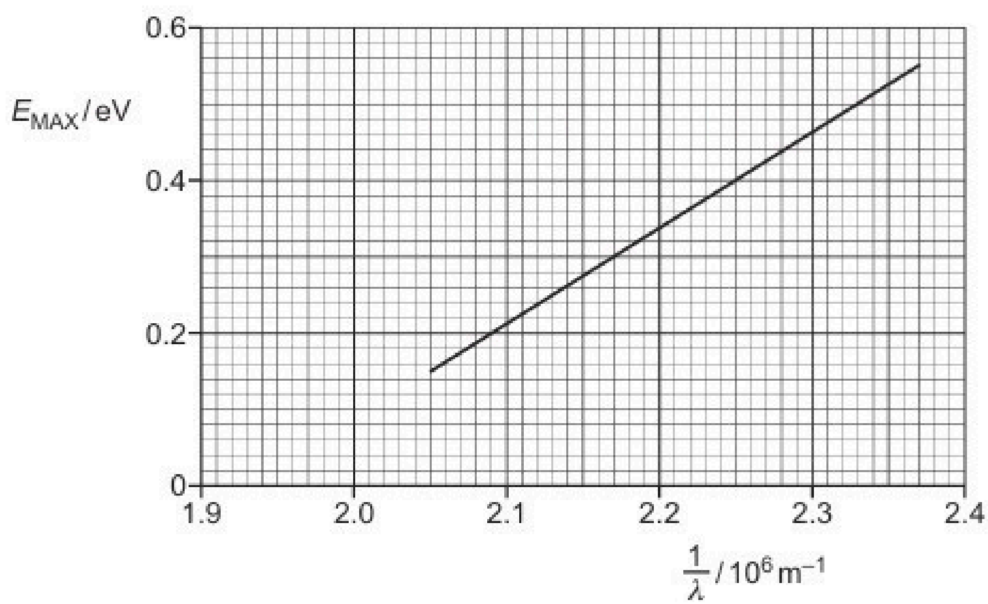

Fig. 11.1

(i) Use Fig. 11.1 to determine the threshold frequency f_0.

f_0 = ... Hz [2]

(ii) Use the gradient of the line on Fig. 11.1 to determine a value for the Planck constant h. Explain your working.

h = ... J s [4]

(c) The electromagnetic radiation is now incident on a metal with a larger work function energy than the metal in **(b)**.

On Fig. 11.1, sketch the variation with $\frac{1}{\lambda}$ of E_{MAX}. [2]

[Total: 10]

44. M/J/21/41/Q12

(a) Electromagnetic radiation of a single constant frequency is incident on a metal surface. This causes an electron to be emitted.

Explain why the maximum kinetic energy of the electron is independent of the intensity of the incident radiation.

....

....

....

....

.... [3]

(b) Ultraviolet radiation of wavelength 250 nm is incident on the surface of a sheet of zinc. The maximum kinetic energy of the emitted electrons is 1.4 eV.

Determine, in eV:

(i) the energy of a photon of the ultraviolet radiation

energy = ... eV [3]

(ii) the work function energy of the surface of the zinc.

energy = ... eV [2]

[Total: 8]

45. M/J/21/42/Q12

(a) State what is meant by a *photon*.

....

....

.... [2]

(b) A stationary nucleus of samarium-157 ($^{157}_{62}Sm$) emits a gamma-ray (γ-ray) photon of energy 0.57 MeV.

Determine, for one γ-ray photon:

(i) its wavelength

wavelength = ... m [2]

(ii) its momentum.

momentum = ... N s [2]

(c) **(i)** Using your answer to **(b)(ii)**, determine the speed of the samarium-157 nucleus after emission of the photon.

speed = ... $m\,s^{-1}$ [2]

(ii) By reference to your answer in **(c)(i)**, explain quantitatively why the speed of the samarium-157 nucleus may be assumed to be negligible compared with the speed of the photon.

....

.... [1]

[Total: 9]

46. O/N/21/41/Q10

(a) State an experimental phenomenon that provides evidence for:

(i) the particulate nature of electromagnetic radiation

.... [1]

(ii) the wave nature of matter.

.... [1]

(b) A particle of matter moves with momentum p.

(i) State the equation that gives the effective wavelength λ of the particle. State the name of any other symbols used.

[2]

(ii) State the name given to the wavelength of the moving particle.

.... [1]

(c) Electrons are accelerated from rest through a potential difference (p.d.) of 4.8 kV.

(i) Show that the final speed of the electrons is $4.1 \times 10^7\,\mathrm{m\,s^{-1}}$.

[2]

(ii) Calculate the effective wavelength of a beam of electrons moving at the speed in **(c)(i)**.

wavelength = ... m [2]

[Total: 9]

47. O/N/21/42/Q9

(a) State what is meant by:

(i) the *photoelectric effect*

....

....

.... [2]

(ii) *work function energy*.

....

.... [1]

(b) A polished calcium plate in a vacuum is investigated by illuminating the surface with light.

It is found that no photoelectric current is produced when the frequency of the light is less than 6.93×10^{14} Hz.

(i) State the name of the frequency below which no photoelectric current is produced.

.... [1]

(ii) Explain how the photon model of electromagnetic radiation accounts for this phenomenon.

....

....

....

.... [3]

(iii) Calculate the work function energy, in eV, of calcium.

work function energy = ... eV [2]

[Total: 9]

48. F/M/22/42/Q8

(a) State the formula for the de Broglie wavelength λ of a moving particle.

State the meaning of any other symbol used.

....

....

.... [2]

(b) Electrons accelerate through a potential difference, pass through a thin crystal and are then incident on a fluorescent screen.

The pattern in Fig. 8.1 is observed on the fluorescent screen.

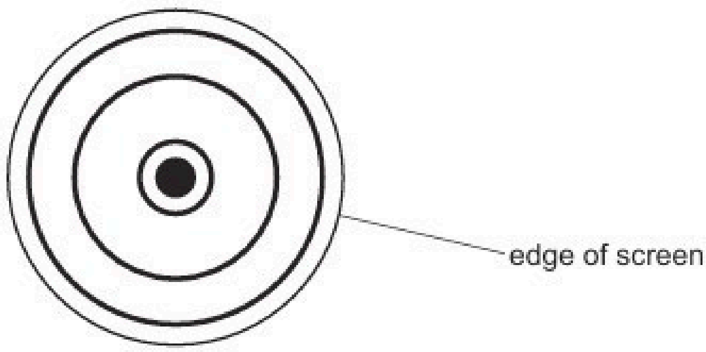

Fig. 8.1 not to scale

(i) State the name of the phenomenon shown by the electrons at the crystal.

.... [1]

(ii) State what this phenomenon shows about the nature of electrons.

....

.... [1]

(iii) Suggest why the thin crystal causes the phenomenon in **(b)(i)**.

....

.... [1]

(iv) The electron is accelerated through a different potential difference. The new pattern observed on the screen is shown in Fig. 8.2.

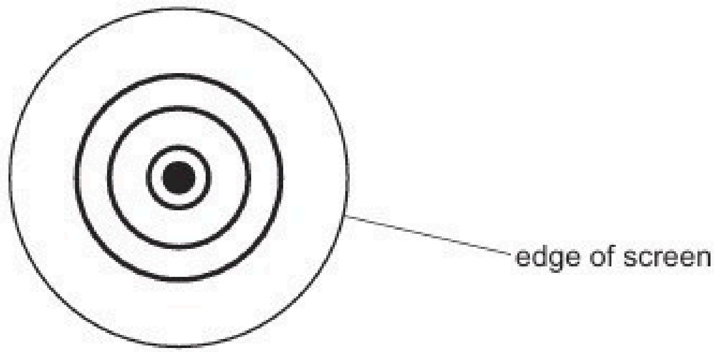

Fig. 8.2 not to scale

State and explain the change that has been made to the potential difference to create the pattern shown in Fig. 8.2.

....

....

....

.... [2]

[Total: 7]

49. M/J/22/41/Q7

(a) State what is meant by a photon.

....

....

.... [2]

(b) Electromagnetic radiation of a varying frequency f and constant intensity I is used to illuminate a metal surface. At certain frequencies, electrons are emitted from the surface of the metal. The variation with f of the maximum kinetic energy E_{MAX} of the emitted electrons is shown in Fig. 7.1.

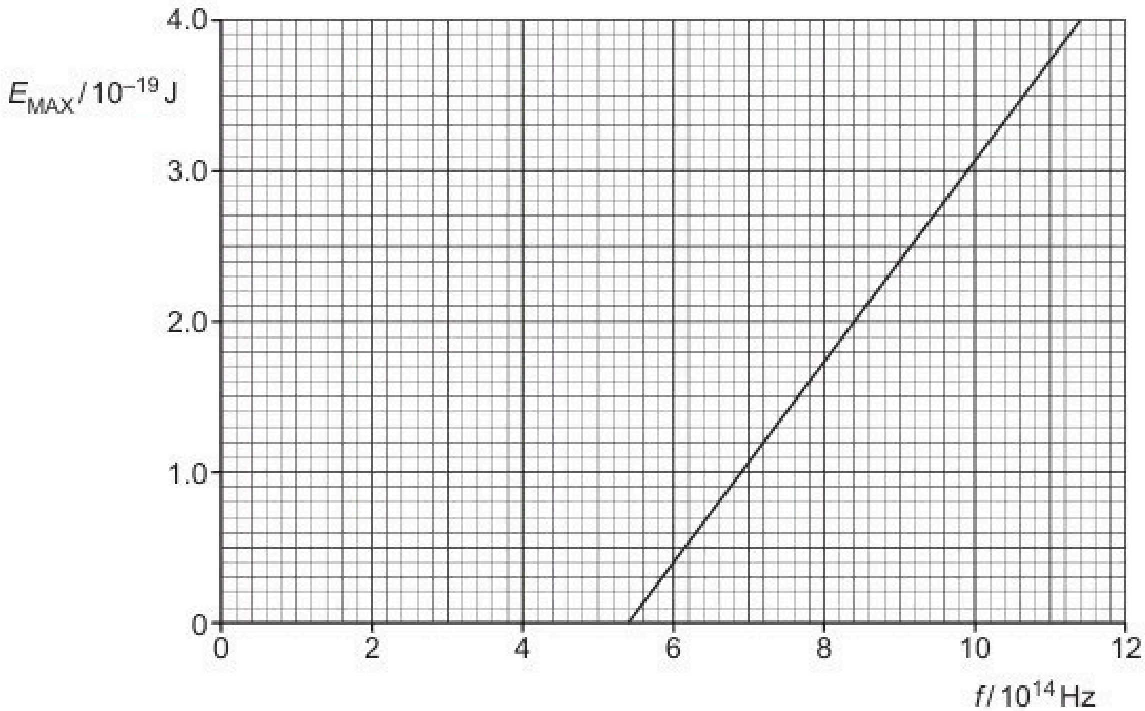

Fig. 7.1

(i) State the name of this phenomenon.

.... [1]

(ii) Describe **three** conclusions that can be drawn from the graph in Fig. 7.1. The conclusions may be qualitative or quantitative.

1 ....

....

2 ....

....

3 ....

....
[3]

(c) The experiment in **(b)** is repeated twice, each time making one change.

State, with a reason, how the graph obtained would compare with Fig. 7.1 when:

(i) a different metal is used, but keeping the intensity I of the radiation the same

....

....

.... [2]

(ii) the same metal is used, but with electromagnetic radiation of intensity $2I$.

....

....

.... [2]

[Total: 10]

50. M/J/22/42/Q8

(a) State **one** piece of experimental evidence for:

(i) the particulate nature of electromagnetic radiation

.... [1]

(ii) the wave nature of matter.

.... [1]

(b) **(i)** Calculate the de Broglie wavelength λ of an alpha-particle moving at a speed of $6.2 \times 10^7\,\mathrm{m\,s^{-1}}$.

λ = ... m [3]

(ii) The speed v of the alpha-particle in **(b)(i)** is gradually reduced to zero.

On Fig. 8.1, sketch the variation with v of λ.

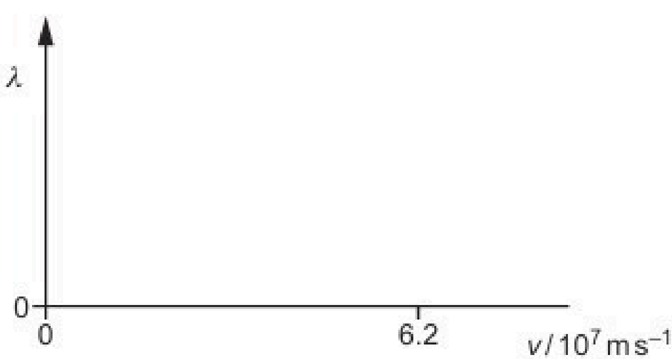

Fig. 8.1

[2]

(c) Suggest an explanation for why people are not observed to diffract when they walk through a doorway.

....

....

.... [1]

[Total: 8]

51. O/N/22/41/Q8

(a) State what is meant by the work function energy of a metal.

....

....

.... [2]

(b) Ultraviolet radiation of frequency 1.36×10^{15} Hz is incident, in a vacuum, on a metal surface. The power of the radiation incident on the surface is 8.36 mW. Photoelectrons are emitted with a maximum kinetic energy of 3.09×10^{-19} J.

(i) Determine the number of photons incident on the surface per unit time.

number per unit time = ... s^{-1} [2]

(ii) Calculate the work function energy Φ of the metal.

Φ = ... J [2]

(c) The frequency of the radiation incident on the surface in **(b)** is increased while the power remains constant.

State and explain the effect of this change on:

(i) the maximum kinetic energy of the photoelectrons

....

....

.... [2]

(ii) the rate of emission of photoelectrons.

....

....

.... [2]

52. O/N/22/42/Q9

(a) Fig. 9.1 shows the visible part of the emission spectrum from hydrogen gas in a laboratory on the Earth. The numbers indicate the wavelength, in nm, represented by each line.

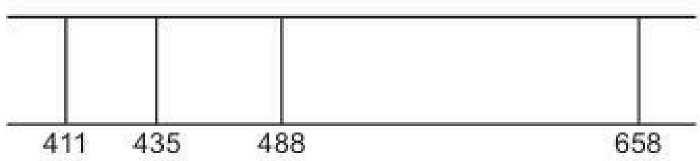

Fig. 9.1

(i) Explain how the emission spectrum provides evidence for the existence of discrete energy levels for the electron in a hydrogen atom.

....

....

....

.... [3]

(ii) Fig. 9.2 shows five of the energy levels in the hydrogen atom. The wavelengths of radiation shown in Fig. 9.1 relate to transitions to the −3.400 eV level in Fig. 9.2.

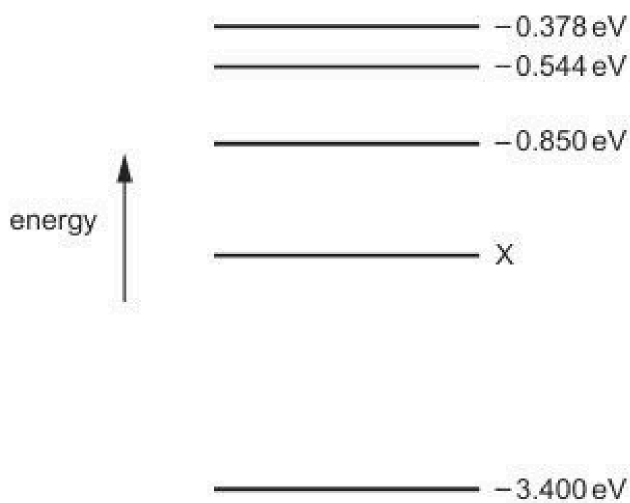

Fig. 9.2 (not to scale)

Show that the energy level X is −1.51 eV.

[3]

(b) The same part of the emission spectrum from hydrogen as in **(a)**, observed in light from stars in a distant galaxy, is shown in Fig. 9.3. The numbers indicate the wavelengths in nm.

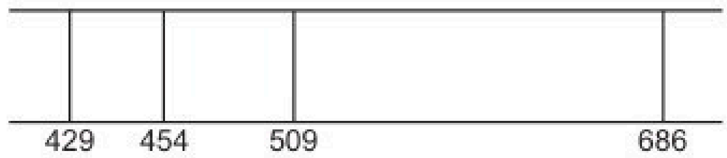

Fig. 9.3

The spectrum shows the same pattern as Fig. 9.1 but with different wavelengths.

(i) State the name of the phenomenon that gives rise to the change in the wavelengths.

.... [1]

(ii) State what this phenomenon shows about the motion of the galaxy.

.... [1]

(iii) Use one of the lines in Fig. 9.1, and the corresponding line in Fig. 9.3, to determine the speed of the distant galaxy relative to the observer.

speed = ... $m\,s^{-1}$ [3]

(c) The galaxy in **(b)** is known to be a distance of 5.7×10^{24} m from the Earth.

Use your answer in **(b)(iii)** to determine a value for the Hubble constant H_0.

H_0 = ... s^{-1} [2]

[Total: 13]

53. F/M/23/42/Q7

(a) A beam of white light passes through a cloud of cool gas. The spectrum of the transmitted light is viewed and contains a number of dark lines.

Explain why these dark lines occur.

....

....

....

....

....

....

....

....

.... [4]

(b) Some energy levels for the electron in an isolated hydrogen atom are illustrated in Fig. 7.1.

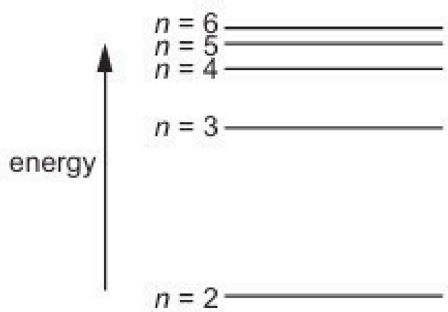

Fig. 7.1

Table 7.1 shows the wavelengths of photons that are emitted in the transitions to n = 2 from the other energy levels shown in Fig. 7.1.

Table 7.1

wavelength / nm
412
435
488
658

The energy associated with the energy level n = 2 is −3.40 eV.

Calculate the energy, in J, of energy level $n = 3$.

energy = ... J [3]

[Total: 7]

54. M/J/23/41/Q7

(a) State what is meant by the de Broglie wavelength.

....

.... [1]

(b) Fig. 7.1 shows a glass tube in which electrons are accelerated through a high p.d. to form a beam that is incident on a thin graphite crystal.

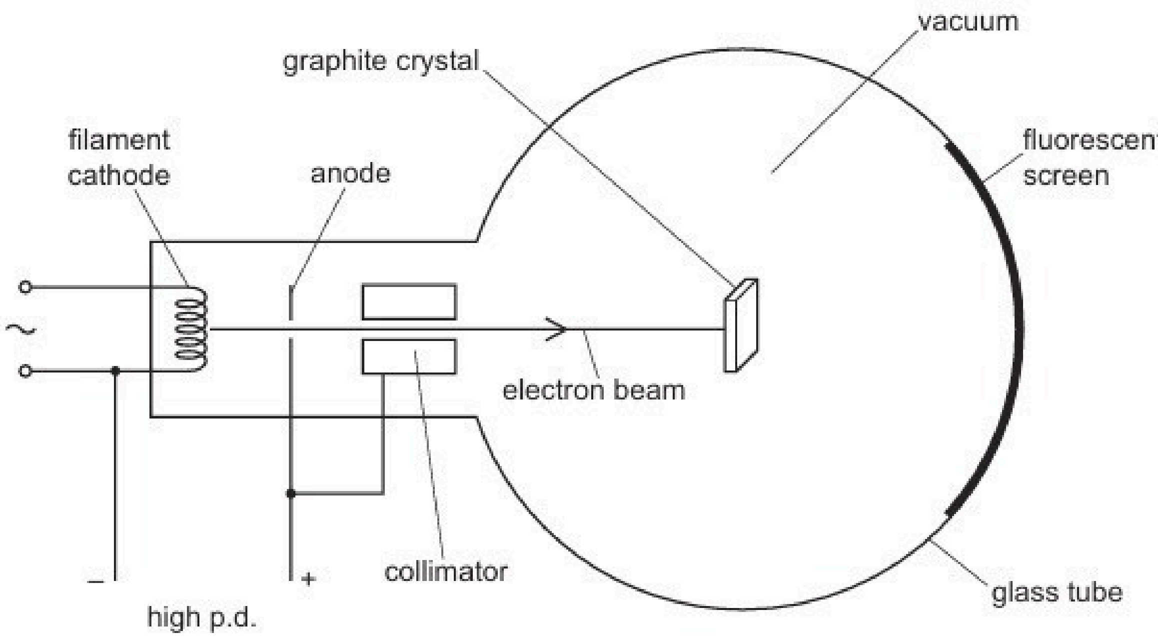

Fig. 7.1 (not to scale)

After passing through the graphite crystal, the electrons reach the fluorescent screen. The screen glows where the electrons strike it.

Fig. 7.2 shows the fluorescent screen viewed end-on, from the right-hand side of Fig. 7.1.

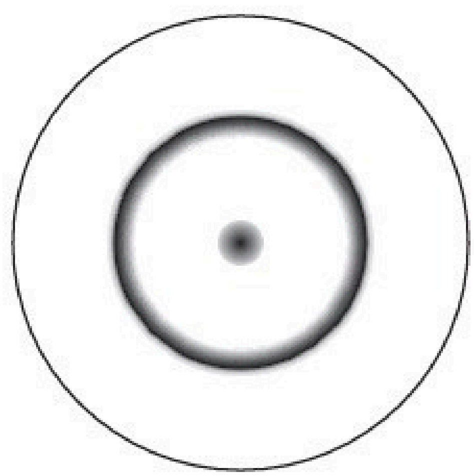

Fig. 7.2

(i) State the name of the phenomenon demonstrated by the pattern shown in Fig. 7.2.

.... [1]

(ii) Explain what can be concluded from the pattern in Fig. 7.2 about the nature of electrons.

....

....

.... [2]

(c) The electrons in **(b)** are now accelerated through a greater potential difference between the cathode and the anode.

(i) On Fig. 7.3, sketch the pattern that is now seen on the fluorescent screen in Fig. 7.1.

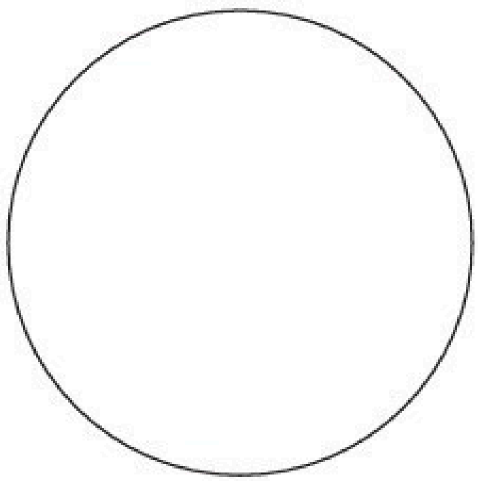

Fig. 7.3

[2]

(ii) Explain, with reference to de Broglie wavelength, the change in the pattern on the fluorescent screen.

....

....

....

....

.... [3]

[Total: 9]

55. M/J/23/42/Q8

Fig. 8.1 shows the lowest four energy levels of an electron in an isolated atom.

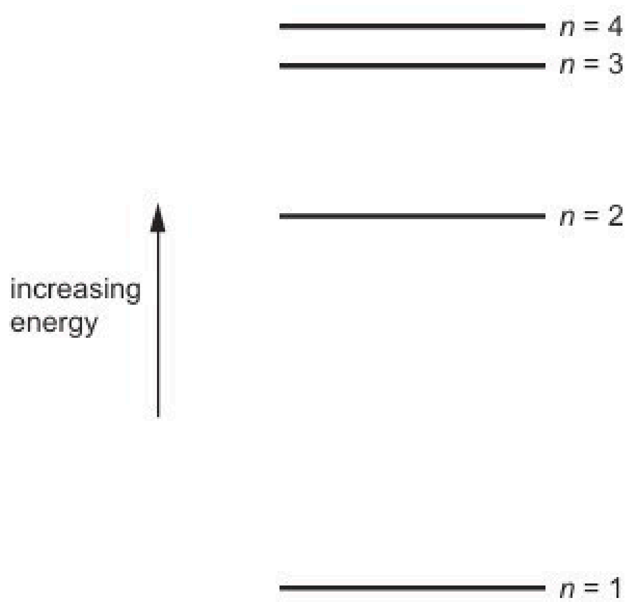

Fig. 8.1

Fig. 8.2 shows the lines in the emission spectrum of the atom that correspond to the transitions of the electron from $n = 3$ to $n = 1$ and from $n = 4$ to $n = 1$.

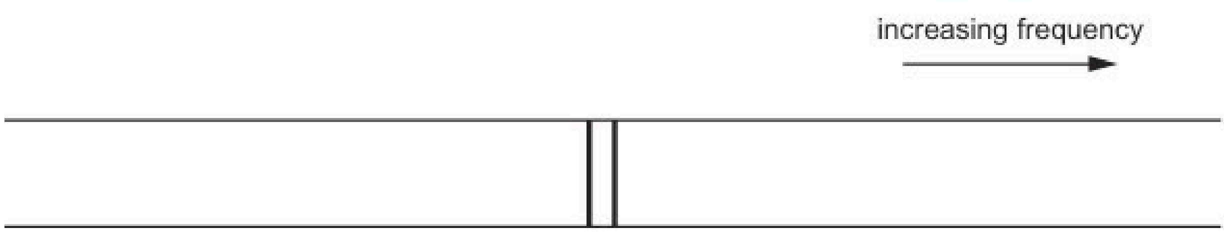

Fig. 8.2

(a) Explain, with reference to photons, why there is a single frequency of electromagnetic radiation that corresponds to each of these transitions.

....

....

.... [2]

(b) **(i)** On Fig. 8.2, draw a line that corresponds to the transition of the electron from $n = 2$ to $n = 1$. Label this line A. [2]

(ii) On Fig. 8.2, draw a line that corresponds to the transition of the electron from $n = 3$ to $n = 2$. Label this line B. [2]

(c) The frequency of radiation represented by line A is f_A.
The frequency of radiation represented by line B is f_B.
The energy of the ground state (n = 1) is E_1.

Determine an expression, in terms of f_A, f_B, E_1 and the Planck constant h, for the energy E_3 of the energy level n = 3.

E_3 = ... [2]

[Total: 8]

56. O/N/23/41/Q8

(a) **(i)** Show that the momentum p of a photon of electromagnetic radiation with wavelength λ is given by

$$p = \frac{h}{\lambda}$$

where h is the Planck constant.

[2]

(ii) Use the expression in **(a)(i)** to show that a photon in free space that has a momentum of 9.5×10^{-28} N s is a photon of red light.

[1]

(b) A beam of red light of intensity $160\,\text{W m}^{-2}$ is incident normally on a plane mirror, as shown in Fig. 8.1. The momentum of each photon in the beam is 9.5×10^{-28} N s.

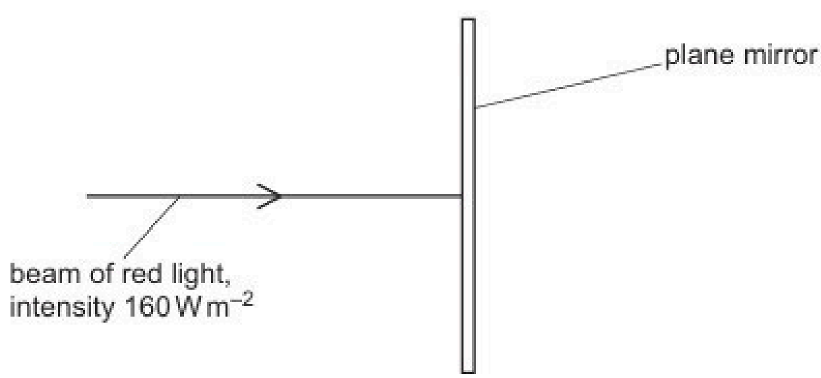

Fig. 8.1

All of the light is reflected by the mirror in the opposite direction to its original path.
The cross-sectional area of the beam is $2.5 \times 10^{-6}\,\text{m}^2$.

(i) Show that the number of photons incident on the mirror per unit time is $1.4 \times 10^{15}\,s^{-1}$.

[2]

(ii) Use the information in **(b)(i)** to determine the pressure exerted by the light beam on the mirror.

pressure = ... Pa [3]

(c) The beam of red light in **(b)** is now replaced with a beam of blue light of the same intensity.

Suggest and explain whether the pressure exerted on the mirror by the beam of blue light is less than, the same as, or greater than the pressure exerted by the beam of red light.

....

....

....

.... [2]

[Total: 10]

57. O/N/23/42/Q8

(a) State what is meant by a photon.

....

....

.... [2]

(b) When the surface of a metal plate is illuminated with electromagnetic radiation, electrons are sometimes emitted from the metal.

(i) State the name of this phenomenon.

.... [1]

(ii) It is observed that this phenomenon occurs only when the frequency of the electromagnetic radiation is greater than a certain minimum value, regardless of the intensity of the radiation.

Explain how this observation provides evidence for the existence of photons.

....

....

....

....

.... [3]

(c) Fig. 8.1 shows the variation of the maximum kinetic energy of the emitted electrons in **(b)** with the frequency of the incident radiation.

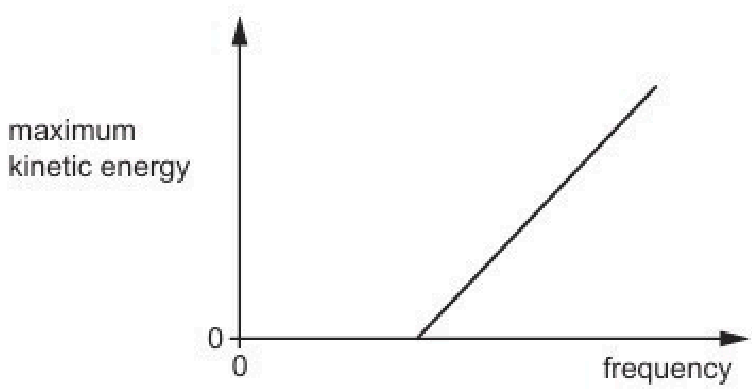

Fig. 8.1

State the name of the quantity represented by:

(i) the gradient of the line in Fig. 8.1

.... [1]

(ii) the *y*-intercept of the extrapolated line in Fig. 8.1.

.... [1]

[Total: 8]